

Bilingual Voice Transcription and Synthesis Preserving Speaker Identity Using NLP and Deep Learning

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August 2026

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A dissertation submitted to the
Department of Computer Science and Informatics,
Faculty of Applied Sciences,
Uva Wellassa University

in partial fulfilment of the requirements for the award of the
Degree of Bachelor of Science Honours in Computer Science and Technology

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DECLARATION

We do hereby declare that the work reported in this dissertation was exclusively carried out by us under the supervision of Mr. E. M. U. W. J. B. Ekanayake and Mr. H. P. D. P. Pathirana. It describes the results of our own independent research except where due reference has been made in the text. No part of this dissertation has been submitted earlier or concurrently for the same or any other degree.

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ACKNOWLEDGEMENT

We wish to record our sincere gratitude to our supervisor, Mr. E. M. U. W. J. B. Ekanayake, and our co-supervisor, Mr. H. P. D. P. Pathirana, of the Department of Computer Science and Informatics, for their guidance throughout this research project. Their insistence that every claim be traced back to a measurement, and their willingness to question results that looked convenient, shaped this dissertation more than any other single influence.

We thank the Head and the academic staff of the Department of Computer Science and Informatics, Faculty of Applied Sciences, Uva Wellassa University, for the research project framework, the proposal and progress review panels, and the feedback given at each stage. The correction of the fundamental-frequency measurement reported in Section 4.9 was a direct consequence of a panel question about whether a reported correlation could be trusted.

This study would not have been possible without the *Dialogs Re-enacted Across Languages* (DRAL) corpus, released into the public domain by Professor Nigel G. Ward and colleagues at the University of Texas at El Paso. A corpus in which the same bilingual speaker performs the same content in two languages is rare, and it is the instrument on which the central measurement of this dissertation depends. We are grateful to the research groups who released Whisper, MarianMT, XTTS-v2, ECAPA-TDNN and SpeechBrain as open models and open source; this project stands on all of them.

Finally, we thank our families and our colleagues in the 2021 intake for their support and patience over the course of this work.

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ABSTRACT

Cross-lingual speech synthesis can now clone a speaker’s voice into a language that speaker never used, but whether it also carries their prosody — the pitch, rhythm and timing that make speech expressive — has rarely been measured, because most corpora contain no recording of what the same speaker actually sounds like in the target language. This study implements a cascaded English–Spanish speech-to-speech translation system (Whisper, MarianMT, XTTS-v2) and evaluates it on the Dialogs Re-enacted Across Languages corpus, in which one bilingual speaker performs matched content in both languages and therefore supplies a human cross-language reference for every generated utterance. Eight hundred and seventy held-out utterances were synthesised in both directions over a speaker-disjoint split, and speaker identity, intelligibility, translation quality and prosody were measured against that reference.

Speaker identity reached 94.5 % (English to Spanish) and 86.1 % (Spanish to English) of the human cross-language ceiling, so most of the identity that is retainable across a language switch survives. Prosody did not follow. Within a speaker, human pitch level transfers at $r = 0.787$, whereas synthesised speech reaches only 0.311 and 0.314, and pitch range collapses from 0.214 to 0.035–0.040. Duration transfers better (0.352–0.610 against 0.870), but raw synthesis runs $1.52\times$ longer than the human target, which a WORLD-based retiming stage corrects inside the accompanying dubbing application. A ridge model predicting target-language prosody from source-language prosody improved on a copy-plus-offset baseline only for pitch range, by 9.5 % and 14.7 % mean absolute error, and no prosody result survives family-wise correction. Fine-tuning MarianMT on the corpus raised BLEU by 2.34 and 3.23, which is register adaptation rather than improved adequacy. The contribution is a calibrated measurement: zero-shot voice cloning preserves identity but not prosody, and the size of that gap is now quantified against a human ceiling.

Keywords: *speech-to-speech translation; speaker identity; cross-lingual prosody; zero-shot voice cloning; automatic dubbing*

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LIST OF ABBREVIATIONS

AdamW	Adam optimiser with decoupled weight decay
API	Application Programming Interface
APA	American Psychological Association (referencing style)
ASR	Automatic Speech Recognition
BLEU	Bilingual Evaluation Understudy
BH	Benjamini–Hochberg (false discovery rate procedure)
chrF	Character <i>n</i> -gram F-score
CI	Confidence Interval
CPU	Central Processing Unit
CVSS	Common Voice-based Speech-to-Speech translation corpus
DRAL	Dialogs Re-enacted Across Languages
DSP	Digital Signal Processing
ECAPA	Emphasized Channel Attention, Propagation and Aggregation
EN	English
ES	Spanish
F0	Fundamental frequency (perceived as pitch)
FDR	False Discovery Rate
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
MAE	Mean Absolute Error
MedAE	Median Absolute Error
MOS	Mean Opinion Score
MPS	Metal Performance Shaders (Apple GPU backend)
MT	Machine Translation
NLP	Natural Language Processing
NMT	Neural Machine Translation
OPUS	Open Parallel Corpus
PESQ	Perceptual Evaluation of Speech Quality
pYIN	Probabilistic YIN fundamental frequency estimator
RMSE	Root Mean Squared Error
RMS	Root Mean Square (signal energy)
S2ST	Speech-to-Speech Translation
SD	Standard Deviation
STOI	Short-Time Objective Intelligibility
TDNN	Time Delay Neural Network
TTS	Text-to-Speech
UWU	Uva Wellassa University
WER	Word Error Rate

WORLD
XTTS

Vocoder-based speech analysis/synthesis system of Morise et al.
Cross-lingual Text-to-Speech (Coqui XTTS-v2)

CHAPTER 1

INTRODUCTION

1.1 Rationale of the Research

Speech carries two kinds of information at once. It carries *what* was said, and it carries *how* it was said — who the speaker is, how fast they talk, where they place emphasis, how their pitch rises at the end of a question. Machine translation systems have become very good at the first and have historically discarded the second. A translated utterance played back in a flat, unfamiliar synthetic voice is understandable, but it is no longer recognisably the same person speaking.

Two developments have changed this. Zero-shot voice cloning systems such as YourTTS (Casanova et al., 2022) and XTTS-v2 (Casanova et al., 2024) synthesise speech in a target language from a few seconds of reference audio by a speaker who never spoke that language; and robust multilingual recognition (Radford et al., 2023) together with openly released translation models (Tiedemann & Thottingal, 2020) make a complete speech-to-speech translation (S2ST) pipeline assemblable on ordinary hardware. The applications are immediate: automatic dubbing of film and educational video, cross-lingual conferencing, and assistive communication.

The claim usually made for such systems is that they *preserve speaker identity across languages*. That claim is normally supported with a cosine similarity between speaker embeddings of the source recording and the synthesised output. The difficulty is that such a number has no natural scale. A cosine of 0.42 could indicate excellent preservation or poor preservation; nothing in the measurement says which, because there is no reference point for how similar the *same person* sounds to themselves when they switch language. Human speakers do not sound identical across languages either — phonetic inventory, rhythm class and habitual pitch range all shift — so part of the loss attributed to the synthesiser may simply be the cost of changing language at all.

A second and larger difficulty concerns prosody. Identity and prosody are routinely

conflated under the single heading of “naturalness”, yet they are separate properties with separate mechanisms. A voice-cloning synthesiser is conditioned on the reference recording for *timbre*. It is not, in general, given the source speaker’s pitch contour, and it therefore invents its own intonation from the text it is asked to read. Whether the resulting prosody bears any relation to what the speaker would actually have done in the target language is an empirical question, and it can only be answered if a recording of that speaker doing precisely that exists.

This dissertation is built around a corpus for which such recordings do exist. The Dialogs Re-enacted Across Languages (DRAL) corpus (Ward, Avila, Rivas, & Marco, 2023) asks bilingual participants to hold a natural conversation in one language and then to re-enact the same exchange, fragment by fragment, in the other. The result is a set of utterance-aligned pairs in which *one* speaker produces both sides. That design turns two otherwise unanswerable questions into measurements: how similar does a speaker sound to themselves across languages, and how much of their prosody carries across. Both become calibrated ceilings against which a synthesis pipeline can be judged.

1.2 Problem Identification

Existing bilingual and multilingual speech synthesis systems produce intelligible cross-lingual speech, but the extent to which they preserve the speaker’s individuality and expressive behaviour remains poorly quantified. Three specific gaps motivate this study.

Uncalibrated identity metrics. Speaker-similarity scores are reported as bare cosines against the source recording. Without a same-speaker cross-language anchor, it is impossible to say what fraction of the achievable similarity a system actually attains, or how much of the shortfall is attributable to the synthesiser rather than to the language change itself.

Unmeasured prosody transfer. Cross-lingual prosody transfer is an active research area (Swiatkowski et al., 2023; Song et al., 2023), but progress is impeded by the scarcity of parallel expressive data. Work in this area consequently reports perceptual scores or within-language proxies rather than a direct comparison against what the same speaker did in the target language. The size of the gap between human and synthetic cross-lingual prosody has therefore not been stated in units that a subsequent system can be measured against.

Confounded reporting. Correlations computed over a pooled set of utterances from many speakers conflate two different things: the tendency of a low-pitched person to remain low-pitched in either language, and the transfer of utterance-level expressive variation. The first is speaker identity; only the second is prosody transfer. Pooled statistics that do not separate them systematically overstate how much prosody a system carries.

A fourth, practical problem emerged during the study and is reported alongside the others. Synthesised utterances are systematically longer than the speech they replace, so a dubbed recording drifts out of synchronisation with the picture within a few utterances unless the generated audio is retimed.

1.3 Research Questions

- RQ1. How much of a speaker’s identity, measured against a same-speaker cross-language ceiling, survives zero-shot voice cloning in an English–Spanish S2ST cascade?
- RQ2. How much of a speaker’s prosody — pitch level, pitch range, duration and speaking rate — transfers across languages when a human does it, and how much when the system does it?
- RQ3. Is the target-language prosody of an utterance predictable from its source-language prosody beyond a trivial copy-plus-offset baseline?
- RQ4. What is the cost, in translation quality, of placing an automatic recogniser in front of the translation model, and does in-domain fine-tuning of the translation model recover it?

1.4 Aim and Objectives

Aim. To develop and rigorously evaluate a bilingual (English–Spanish) voice transcription and synthesis system that translates speech between the two languages while preserving the speaker’s vocal identity, and to quantify how much of the speaker’s prosody the system carries, using a same-speaker parallel corpus as the reference standard.

Objectives. The three objectives registered in the approved research proposal, restated

in the terms in which they were finally executed, are:

- O1.** To design and implement a bidirectional voice-to-voice translation framework for English and Spanish that preserves the speaker’s voice characteristics, using pre-trained ASR, NMT and voice-cloning TTS components, and to measure identity preservation against a human cross-language reference.
- O2.** To analyse the prosodic relationship between the two languages for the same speaker, to measure how much of that relationship the system reproduces, and to test whether target-language prosody can be predicted from source-language prosody.
- O3.** To integrate the synthesised speech with the original video, applying temporal alignment so that the dubbed track keeps the timing of the original, and to deliver this as a working application.

A fourth activity, not registered as an objective but required to answer RQ4, was the fine-tuning of the translation component on the in-domain conversational data and its comparison against the pre-trained baseline.

1.5 Scope and Delimitations

The study is deliberately bounded, and the boundaries are stated here so that the results are read correctly.

- **Two languages, both directions.** English and Spanish only, evaluated in both $EN \rightarrow ES$ and $ES \rightarrow EN$.
- **One corpus.** All experimental data come from DRAL. The pre-trained components carry knowledge from their own training corpora, which were not modified.
- **Objective evaluation only.** No listening study was conducted. Every result in Chapter 4 is an instrumental measurement. Mean opinion scores, and the PESQ and STOI metrics named in the proposal, are not reported; Section 4.12 explains why the latter two are not defined for this task.
- **Analysis, not control.** The prosody model developed in this study *predicts* target-language prosody; it does not condition the synthesiser. Closing the loop is identified as future work.
- **Duration-matched dubbing, not lip synchronisation.** The application aligns

each synthesised segment to the duration of the segment it replaces. It does not perform phoneme-level lip synchronisation, speaker diarisation, or separation of background music.

1.6 Expected Outcomes and Contributions

The dissertation delivers:

1. A reproducible bidirectional English–Spanish S2ST pipeline with zero-shot voice cloning, evaluated on 870 held-out utterances over a speaker-disjoint split, every failure retained.
2. A *calibrated* identity result — speaker similarity as a percentage of the human same-speaker cross-language ceiling — and a quantification of the prosody transfer gap reported within-speaker, so that identity is not counted twice as prosody.
3. A negative result with a stated mechanism: target-language prosody is predictable beyond a copy-plus-offset baseline only for pitch *range*, and the learned coefficient explains why.
4. A methodological finding: correlating pitch in hertz without a plausibility gate understated the correspondence by roughly a factor of seven (Section 4.10).
5. A working web application that dubs an uploaded video into the other language in the original speaker’s voice, with each segment retimed to its original slot.

1.7 Structure of the Dissertation

Chapter 2 reviews cascaded and direct S2ST, identity-preserving synthesis, cross-lingual prosody transfer and the evaluation methodology on which this study depends, and identifies the gap it addresses. Chapter 3 describes the corpus, the partitioning, the architecture, every measurement instrument and each experiment. Chapter 4 reports and discusses the results in the same order, and Chapter 5 evaluates them against the objectives and sets out the next steps.

CHAPTER 2

LITERATURE REVIEW

2.1 Cascaded and Direct Speech-to-Speech Translation

Speech-to-speech translation has been approached in two ways. The *cascaded* approach chains automatic speech recognition, machine translation and speech synthesis, so that each stage can be built, replaced and measured independently. The *direct* approach trains a single model to map source speech to target speech, either through an intermediate spectrogram (Jia, Ramanovich, Remez, & Pomerantz, 2022) or through discrete acoustic units learned without text (Lee et al., 2022; Inaguma et al., 2023).

The trade-off is well documented. Direct models avoid the propagation of recognition errors into translation and can, in principle, carry paralinguistic information that text discards; cascades remain competitive in translation quality, are far cheaper to assemble, and are interpretable at every stage. Sperber and Paulik (2020) argue that most of the theoretical advantages claimed for end-to-end systems have yet to be realised in practice, and Bentivogli et al. (2021) find the remaining quality difference on standard benchmarks to be small. For a study whose object is *measurement* rather than architecture, the cascade has a decisive advantage: the intermediate text makes it possible to attribute an observed error to a specific stage, and to quantify what each stage costs. That is the argument for the design adopted here.

2.2 Automatic Speech Recognition

Radford et al. (2023) trained the Whisper family on approximately 680,000 hours of weakly supervised multilingual audio, showing that scale alone yields zero-shot robustness competitive with supervised systems. Whisper also emits segment-level timestamps, which matters here: dubbing continuous video requires knowing where each utterance sits on the timeline, not merely what was said. Its known weaknesses

— disfluency, overlapping talk, short backchannels — are exactly the conditions of the conversational corpus used here, and the resulting error rate is reported rather than suppressed.

2.3 Neural Machine Translation and Length Control

Transformer-based translation (Vaswani et al., 2017) is standard. The Marian toolkit (Junczys-Dowmunt et al., 2018) and the OPUS-MT release (Tiedemann & Thottingal, 2020) provide small, fast, openly licensed bilingual models trained on OPUS — books, subtitles and institutional documents. Their register is written text, a documented mismatch for spontaneous dialogue; in-domain fine-tuning is the standard remedy and is examined empirically in Chapter 4.

For dubbing specifically, translation quality is not the only requirement: the translated utterance must also be *speakable in the time available*. Federico et al. (2020) decompose automatic dubbing into translation, prosodic alignment and audio rendering, and Lakew et al. (2021) show that verbosity can be controlled at translation time by conditioning on a desired length class. This literature anticipates the duration problem reported in Section 4.9 and points to its proper solution.

2.4 Speaker-Identity-Preserving Speech Synthesis

Neural text-to-speech moved from Tacotron 2 with a neural vocoder (Shen et al., 2018; Kong, Kim, & Bae, 2020) to fully end-to-end variational models (Kim, Kong, & Son, 2021). Multi-speaker synthesis was made zero-shot by conditioning the synthesiser on an embedding produced by a separately trained speaker-verification network (Jia et al., 2018): because the encoder generalises to unseen speakers, a few seconds of reference audio is enough to clone a voice. YourTTS (Casanova et al., 2022) extended this to the multilingual case, and XTTS-v2 (Casanova et al., 2024) scales the idea to seventeen languages with short reference clips. Voice conversion research reaches similar capability from another direction, by disentangling content from speaker information in an autoencoder (Qian, Zhang, Chang, Yang, & Hasegawa-Johnson, 2019).

Two properties of this family matter for the present study. First, conditioning is on *timbre*: the speaker embedding is trained for verification, and it deliberately discards utterance-

level variation. Second, nothing in the architecture supplies the source utterance’s pitch contour to the decoder. The synthesiser must therefore infer intonation from the target text alone. This is the mechanism behind the result reported in Section 4.6, and it is a property of the design rather than a defect of the implementation.

2.5 Prosody Modelling and Cross-Lingual Prosody Transfer

Within a single language, prosody transfer is comparatively mature. Skerry-Ryan et al. (2018) learn a latent prosody embedding from a reference recording and show that conditioning Tacotron on it reproduces the reference’s prosody with fine time detail, even across speakers. Global Style Tokens (Wang et al., 2018) obtain interpretable style control without prosody labels, and Yang (2025) surveys emotional synthesis and voice conversion in the same tradition. Reviews of multilingual voice-to-voice translation (Ambadekar, Dhokiya, Gandhi, & Haria, 2025) and of multi-feature multilingual synthesis (Nuthakki et al., 2023) converge on the same conclusion: translation, synthesis and speaker-preserving components must be integrated, and prosody is the component most often lost.

Cross-lingual prosody transfer is markedly harder, and the obstacle is data. Transferring prosody from a source-language recording to target-language speech requires supervision about what the target-language rendition should sound like, and parallel expressive recordings of the same content in two languages are rare. Song et al. (2023) address this with zero-shot style transfer inside a direct S2ST model. Peng et al. (2024) preserve speaker style in a textless speech language model. The Seamless system (Seamless Communication et al., 2023) adds explicit expressivity and streaming to large-scale multilingual translation. Closest to the present work, Swiatkowski et al. (2023) build a prosody representation for machine dubbing that is transferable across both language and speaker, using a variational reference encoder with a noise modelling module for field recordings — a design necessitated precisely by the absence of clean parallel expressive data.

The corpus used in this dissertation removes that obstacle for measurement purposes. Ward et al. (2023) designed DRAL to support “machine learning of cross-language prosodic mappings”: participants re-enact their own utterances in the other language, so both sides of a pair come from the same speaker performing the same communicative act. What is a scarcity problem for *training* an expressive dubbing model is not a scarcity problem for *evaluating* one.

2.6 Evaluation of Cross-Lingual Speech Generation

Translation quality is measured with BLEU (Papineni, Roukos, Ward, & Zhu, 2002) and chrF (Popović, 2015) through a standardised implementation (Post, 2018), and recognition with word error rate. Speaker similarity is a cosine between speaker-verification embeddings; ECAPA-TDNN (Desplanques, Thienpondt, & Demuynck, 2020), trained on VoxCeleb (Nagrani, Chung, & Zisserman, 2017) and distributed through SpeechBrain (Ravanelli et al., 2021), is the current standard. Prosody is described with fundamental frequency, energy, duration and speaking rate, F0 being estimated by pitch trackers such as pYIN (Mauch & Dixon, 2014) in librosa (McFee et al., 2015).

Three methodological weaknesses recur and are addressed explicitly in Chapter 3. Speaker-similarity cosines are reported without a same-speaker cross-language ceiling, so their scale is undefined. Intrusive signal metrics such as PESQ and STOI are proposed although they require a time-aligned reference of *the same utterance*, which cannot exist when the output is in another language. And correlations are reported pooled across speakers, counting stable between-speaker differences as evidence of transfer; reporting the within-speaker component alongside is the standard correction, adopted here.

2.7 Research Gap

The literature establishes that zero-shot cloning preserves timbre, that within-language prosody transfer is achievable, and that cross-lingual prosody transfer is limited chiefly by parallel expressive data. What it does not establish is a calibrated statement of *how much* identity and *how much* prosody a standard, openly available cascade actually carries when the human answer for the same speaker and the same content is available for comparison.

This dissertation fills that gap. It uses a same-speaker parallel corpus as a measuring instrument: identity is expressed as a fraction of the human cross-language ceiling; prosody transfer is expressed against the human within-speaker correspondence for the same features; and the predictable component of cross-lingual prosody is isolated by testing a supervised mapping against baselines that a useful model must beat. The contribution is therefore methodological and empirical rather than architectural, and its value is that it states, in reusable units, the size of the gap that expressive dubbing systems set out to close.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study follows a quantitative, experimental design within a positivist frame: a working system is constructed, and its behaviour is characterised by instrumental measurement on data it has never seen. Four experiments are reported.

- E1. Synthesis experiment.** The complete cascade is run over the held-out test split in both directions, and recognition accuracy, translation quality, speaker identity, intelligibility and prosody are measured for every utterance (Sections 3.3–3.4).
- E2. Corpus study.** The cross-lingual prosodic relationship between English and Spanish is characterised for the same human speaker over the whole corpus (Section 3.5).
- E3. Prosody prediction experiment.** A supervised mapping from source-language to target-language prosody is fitted on the training split, selected on development data and tested once, against three baselines (Section 3.6).
- E4. Fine-tuning experiment.** The translation component is adapted to the in-domain conversational data and compared against its pre-trained baseline under identical decoding (Section 3.7).

A fifth deliverable, the dubbing application (Section 3.8), is engineering rather than experiment; it realises Objective 3 and is described because it is a research output and because the retiming stage it contains follows directly from a measured result.

The design has one defining property. Because every test utterance has a matching human recording of *the same speaker performing the same content in the target language*, each measurement is made against a human reference rather than in the abstract. That is what converts an uninterpretable similarity score into a percentage of an achievable ceiling.

3.2 Materials: the DRAL Corpus

All experimental data come from the 16 kHz release of the Dialogs Re-enacted Across Languages corpus (Ward et al., 2023), obtained from the University of Texas at El Paso and placed in the public domain by its authors. In the collection protocol, bilingual participants hold a spontaneous conversation in one language; the recording is then segmented, and the same participants re-enact each fragment in the other language, aiming to reproduce not only the content but the manner of the original. Both sides of a pair therefore come from one person performing one communicative act twice.

Only the `fragments-short` portion of the release was used. The `fragments-long` and `recordings` portions contain longer or unsegmented versions of the same material and carry no additional pair structure, and the Japanese material in the release is out of scope. Pairs were resolved through the official `trans_id` links in `metadata/fragments-short.csv` rather than by substituting the language prefix in a file-name, because a number of valid pairs use different conversation or channel identifiers. Human transcripts were merged from the official `EN_transcripts.txt` and `ES_transcripts.txt` files; where a transcript is absent the field is left empty and the affected metric is reported as unavailable, never imputed from recogniser output. Table 3.1 records the resulting inventory, which was validated locally rather than taken from the documentation.

Table 3.1: Inventory of the DRAL 16 kHz `fragments-short` release as validated locally. All figures were produced by `bvt inspect-dral` and `bvt build-manifest`.

Property	Value
Linked English–Spanish fragment pairs	2,893
Audio files opened successfully	5,786
Total speech duration	4.27 hours
Files at 16 kHz	5,784
Unique speaker identifiers	96
Pairs performed by one speaker	2,834
Pairs with differing linked speaker identifiers	59
Non-empty English human transcripts	2,871
Non-empty Spanish human transcripts	2,881

3.2.1 Why This Corpus and Not the Ones Named in the Proposal

The approved proposal named Mozilla Common Voice (Ardila et al., 2020) and the CVSS speech-to-speech corpus (Jia, Ramanovich, Wang, & Zen, 2022). Both were examined and set aside, for one reason. In Common Voice the English and Spanish recordings come from different people, so there is no same-speaker cross-language reference. In CVSS the target-language speech is itself synthesised by a text-to-speech system, so comparing generated speech against it would compare one synthesiser with another rather than with a human. Neither corpus can supply the human ceiling on which the central measurement of this study depends. DRAL can, because one bilingual person performs both sides. The cost of the choice is scale — 4.27 hours against thousands — and it is accepted deliberately: this study measures rather than trains, and measurement needs a valid reference more than it needs volume.

Two files, `EN_072_7.wav` and `ES_072_7.wav`, are stored at 44.1 kHz despite the name of the release; they are resampled at read time for feature extraction and speaker embedding, and the raw data are not modified. One line of `ES_transcripts.txt` lacks a field delimiter; the parser recovers the annotator code and emits a warning rather than editing the source file.

3.2.2 Speaker-Disjoint Partitioning

The corpus was partitioned into training, development and test sets in the proportions 70/15/15 with random seed 498. A conventional random split would be invalid here, because a speaker appearing in both training and test data would let the evaluation reward memorised voice characteristics rather than generalisation. Assignment therefore operates on *speakers*, not on pairs.

Because 59 pairs link fragments whose English and Spanish sides carry different speaker identifiers, the two identifiers of such a pair must never be separated. The splitter first builds a graph in which an edge joins the two speaker identifiers of every pair, computes the connected components of that graph, and assigns whole components to splits. This guarantees that no speaker, and no linked group of speakers, appears on both sides of the train/test boundary; the property is asserted programmatically after assignment. The resulting split contains 2,022 training pairs, 436 development pairs and 435 test pairs. The test split, on which all reported system results are computed, contains 14 speakers.

3.3 System Architecture

The system is a cascade of four pre-trained neural components, illustrated in Figure 3.1. Three participate in generation; the fourth is used only for measurement.

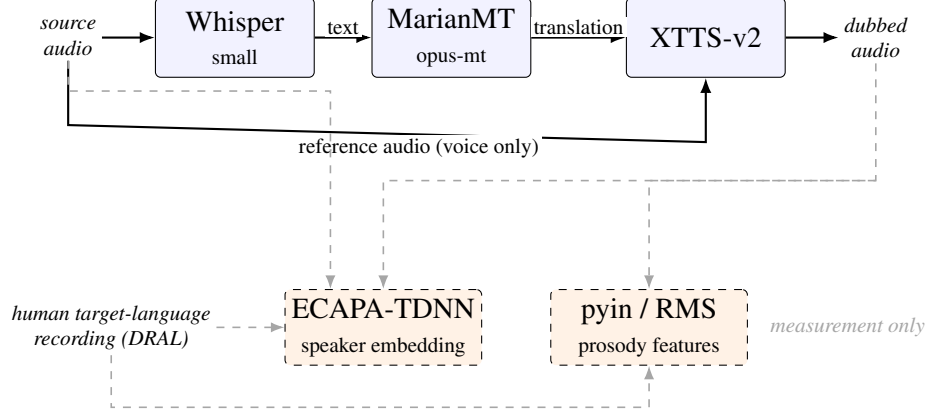


Figure 3.1: Architecture of the cascaded English–Spanish speech-to-speech translation system. Solid arrows are the generation path; the dashed lower layer is measurement only and never influences generation. XTTS-v2 is conditioned on the source recording for voice, but is not given the source pitch contour — a design property that Section 4.6 shows to be consequential.

Recognition. `Whisper small` (Radford et al., 2023) transcribes the source recording, with the source language supplied explicitly rather than auto-detected, so that a recognition error cannot silently become a language-identification error.

Translation. `MarianMT` (Junczys-Dowmunt et al., 2018) in the two `OPUS-MT` bilingual configurations `Helsinki-NLP/opus-mt-en-es` and `Helsinki-NLP/opus-mt-es-en` (Tiedemann & Thottingal, 2020) translates the transcript, decoding with a maximum of 512 new tokens.

Synthesis. `XTTS-v2` (Casanova et al., 2024) synthesises the translation in the target language, conditioned on the *source* recording as the voice reference. The synthesiser receives the target text and the reference waveform; it does not receive the source pitch contour, its energy envelope or its timing.

Measurement. `ECAPA-TDNN` (Desplanques et al., 2020), distributed as `speechbrain/spkrec-ecapa-voxceleb` (Ravanelli et al., 2021), produces speaker embeddings. It generates nothing; it is an instrument.

Determinism is enforced per utterance. The synthesis seed is derived as the first four bytes of `sha256(pair_id | direction)` reduced modulo $2^{31} - 1$, and the framework’s random number generator is seeded with it immediately before synthesis. Python’s

built-in `hash` is salted per process and is deliberately not used. Every row of the results file records the seed that produced it, so any single utterance can be regenerated exactly.

Failures are treated as data. If any stage raises, the row is written with `status = failed`, the failing stage and the exception message, and it remains in the results file and in the reported failure rate. No row is silently dropped.

3.4 Evaluation Instruments

3.4.1 Recognition Accuracy

Word error rate is computed between the human transcript and the Whisper output using `jiwer`, under a fixed normalisation applied identically to reference and hypothesis: lower-casing, punctuation removal, whitespace collapsing and trimming. Both a corpus-level WER (total edits divided by total reference words) and a mean over utterances are reported, because the two answer different questions and diverge when short utterances are common. Utterances without a human transcript are excluded from this metric only.

3.4.2 Translation Quality

Corpus BLEU (Papineni et al., 2002) and chrF (Popović, 2015) are computed with `sacrebleu` (Post, 2018) against the human transcript of the paired target-language recording. That reference deserves a caution which is repeated wherever the resulting scores appear: the DRAL target side is a *re-enactment*, not a literal translation, so the reference is a legitimate rendition of the same communicative content but not a sentence-level gloss. The absolute scores are therefore lower than the same models achieve on written parallel corpora, and only differences measured against the same reference are meaningful.

3.4.3 Speaker Identity and the Human Ceiling

Speaker similarity is the cosine between ECAPA embeddings of two recordings, computed after down-mixing to mono and resampling to 16 kHz. Two quantities are computed for every utterance:

$$S_{\text{system}} = \cos(e(\text{source recording}), e(\text{synthesised output})), \quad (3.1)$$

$$S_{\text{human}} = \cos(e(\text{source recording}), e(\text{human target-language recording})), \quad (3.2)$$

where $e(\cdot)$ denotes the ECAPA embedding. S_{human} is the *ceiling*: it is how similar the same person sounds to themselves when they switch language, measured on real human recordings. Identity preservation is reported as the ratio $100S_{\text{system}}/S_{\text{human}}$. This anchor is available only because DRAL is same-speaker parallel; in a corpus without that property, S_{system} would have no scale.

Both quantities are averaged first within each speaker and then across speakers. The test split is markedly unbalanced — one speaker contributes 118 pairs and another two — so a plain mean over rows would weight the result towards whoever happened to be recorded most. The speaker-level mean gives every speaker equal weight and is the statistic reported throughout; row-level means are given alongside it in Appendix C for completeness.

3.4.4 Intelligibility of Synthesised Speech

The synthesised output is re-transcribed with Whisper and compared against the text that was sent to the synthesiser. This isolates the synthesiser: a high value means the generated waveform is hard to recognise, independently of whether the translation was any good. Rows are filtered on the *reference* being present, never on the hypothesis. An empty re-transcription means the audio was unintelligible, which is a total failure scoring 1.0, not a missing observation; excluding such rows would make the corpus figure look better than it is.

3.4.5 Prosodic Features

Four acoustic descriptors are extracted from every recording with librosa (McFee et al., 2015): fundamental frequency by the probabilistic pYIN estimator (Mauch & Dixon, 2014), frame-level root-mean-square energy, total duration, and speaking rate as words per second from the corresponding text.

Two distinct frequency bands are used, and keeping them distinct is essential to the correctness of the result.

- The **search band**, 65–1000 Hz, is the range given to pYIN. It must be wide. pYIN’s voiced/unvoiced decision depends on the number of frequency candidates available, so narrowing the band lowers the aggregate voiced probability and the estimator stops detecting voicing at all on quiet recordings. This was measured, not assumed (Section 4.10).
- The **plausibility band**, 60–400 Hz, is applied downstream as a data-quality gate. A wide search band admits occasional octave doublings; a median F0 outside the plausible adult speaking range is a tracking failure rather than a voice, and such observations are excluded before any statistic is computed.

All F0 statistics are computed in **semitones** relative to a 100 Hz reference,

$$s = 12 \log_2 \left(\frac{f}{100} \right), \quad (3.3)$$

where f is the measured frequency in hertz, because pitch perception is logarithmic. A correlation computed in hertz is dominated by whichever values lie furthest from the mean, and a single octave-doubled observation can destroy it; the consequences of getting this wrong are reported in Section 4.10. Throughout, *F0 level* denotes the mean semitone value of an utterance and *F0 range* its within-utterance standard deviation in semitones.

3.4.6 Metrics Deliberately Not Used

The approved proposal named PESQ and STOI. Both are intrusive metrics: they compare a degraded signal against a time-aligned clean recording of *the same utterance*. In cross-lingual synthesis no such reference can exist, because the output is in a different language and contains different words of different durations. Reporting them would mean either fabricating an alignment or comparing unrelated signals. They are therefore not reported, and the reason is stated rather than left as an omission. A mean opinion score study was also planned; it was not conducted within the available time and is recorded as a limitation in Section 4.12 and as the first item of future work.

3.5 Corpus Study: Cross-Lingual Prosody in Human Speech

The corpus study characterises the relationship between the English and Spanish prosody of the *same* speaker, over the whole corpus rather than the test split, and supplies the human reference against which the system is judged.

3.5.1 Quality Gate

Both sides of a pair must yield a usable measurement, so a pair is retained only if, on *both* sides: F0 was estimated at all; the voiced-frame ratio is at least 0.30; duration lies between 0.4 and 20 s; and mean F0 lies inside the plausibility band. Analysis is further restricted to pairs performed by a single speaker, since the research question is about one person's behaviour in two languages. The gate is a criterion on the *measurement*, applied before any outcome is computed, and every count it removes is reported in Table 4.1.

3.5.2 Within- and Between-Speaker Decomposition

A Pearson correlation computed over utterances pooled from many speakers conflates two effects. If low-pitched people are low-pitched in both languages, a strong pooled correlation follows from speaker identity alone, with no prosody transfer whatsoever. Every correlation in this dissertation is therefore reported in three forms:

- the **pooled** correlation over all utterances;
- the **between-speaker** correlation, computed over one point per speaker, namely their mean on each side;
- the **within-speaker** correlation, computed after centring both variables inside each speaker, which removes stable speaker differences and leaves only utterance-level variation.

The within-speaker figure is the one that answers the research question, and it is the figure quoted in the discussion. The share of total variance attributable to speaker differences is reported alongside, from a one-way decomposition $SS_{\text{between}} / (SS_{\text{between}} + SS_{\text{within}})$.

3.6 Predicting Target-Language Prosody

The corpus study measures the correspondence; this experiment asks whether it can be *exploited*. For each direction and each of three targets — target F0 level, target F0 range, and $\log(\text{target duration}/\text{source duration})$ — a model predicts the target value from six source-side features: F0 level, F0 range, log duration, voiced ratio, word count and speaking rate.

3.6.1 Baselines

A model is only interesting if it beats what can be obtained without one. Three baselines are fitted on the training split alone:

- B0** *copy source* — predict the source value unchanged (and zero for the log duration ratio).
- B1** *train mean* — predict the training-set mean of the target, ignoring the source entirely.
- B2** *copy plus offset* — predict the source value plus a single global scalar, the mean residual on the training split. This is the baseline to beat: it captures any systematic language-level shift with one number.

3.6.2 Ridge Regression on the Residual

The model is ridge regression (Hoerl & Kennard, 1970) in closed form, with standardised inputs and an unpenalised intercept, fitted to the *residual from copy-source* rather than to the absolute target:

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top (\mathbf{y} - \mathbf{c}), \quad \hat{y} = c + \mathbf{x}^\top \hat{\mathbf{w}} + b, \quad (3.4)$$

where $\mathbf{X}$ holds the standardised source features, $\mathbf{y}$ the targets, $\mathbf{c}$ the copy-source prediction, λ the regularisation strength and b the intercept. Fitting the residual makes the model strictly *nest* B2: as $\lambda \rightarrow \infty$ the weights vanish and the prediction becomes copy-plus-mean-residual, which is exactly B2. The model can therefore only improve on B2 up to estimation noise, which removes an entire class of misleading comparison.

λ is selected on the development split from the grid $\{10^{-3}, 10^{-2}, \dots, 10^6\}$ using the one-standard-error rule: the largest λ whose development error is within one standard error of the best. Plain minimisation is unstable on a few-hundred-row development set, and because the model nests B2, an unlucky small λ can make it appear worse than the trivial baseline for reasons that are pure selection noise. Preferring the most-shrunk admissible λ biases towards the baseline, which is the conservative choice. The test split is scored once.

3.6.3 Statistical Inference with Speaker Clustering

The test split holds roughly 357 usable pairs but only 14 speakers, and errors are correlated within a speaker. Treating each recording as an independent observation would overstate significance. Two clustered procedures are reported instead of the naive per-recording test: a Wilcoxon signed-rank test (Wilcoxon, 1945) on *per-speaker mean absolute errors* ($n = 14$), and a bootstrap confidence interval for the difference in mean absolute error that resamples *speakers* rather than recordings (Efron & Tibshirani, 1993), with 2,000 draws at seed 498. Because six models are fitted, the family-wise position is reported explicitly using Bonferroni and Benjamini–Hochberg (Benjamini & Hochberg, 1995) corrections rather than left implicit.

Two properties of the design produce results that look like defects and are not, so they are stated in advance. For the log duration ratio, B1 and B2 are mathematically identical, since copy-source is zero and copy-plus-offset reduces to the training mean; the comparison between them is therefore exactly null. And the errors of B0, B1 and B2 are symmetric between directions by construction, because exchanging source and target mirrors the error distribution; only the ridge rows genuinely differ between the two directions.

3.7 Fine-Tuning the Translation Component

The pre-trained translation models were trained on written text, whereas the corpus is spontaneous bilingual conversation containing disfluencies, fillers and regional lexis. This experiment measures what in-domain adaptation is worth. Each direction was fine-tuned independently from the pre-trained checkpoint on the training split only, with the settings in Table 3.2.

Table 3.2: Fine-tuning configuration. Decoding settings are identical for the pre-trained and fine-tuned systems, so the comparison isolates the effect of adaptation.

Setting	Value
Starting checkpoints	<code>Helsinki-NLP/opus-mt-en-es, opus-mt-es-en</code>
Training pairs	1,998 per direction (training split only)
Development pairs	429
Test pairs	432 (scored once)
Optimiser	AdamW (Loshchilov & Hutter, 2019), learning rate 2×10^{-5}
Batch size	8, gradient clipping at 1.0
Maximum sequence length	128 tokens
Epochs run	4; checkpoint selected on development loss
Random seed	498
Hardware	Apple M1 GPU via Metal Performance Shaders
Decoding (both systems)	beam size 4, maximum 128 new tokens

The learning rate is deliberately small. A large rate would overwrite the general translation knowledge acquired from OPUS, and 2×10^{-5} is roughly an order of magnitude below typical from-scratch rates, which is the standard range for adapting a pre-trained transformer. Only the translation component was adapted; the reasoning for not fine-tuning the other components is given in Section 4.8.1.

3.8 Dubbing Application

Objective 3 requires the synthesised dialogue to be integrated with the original video. The research pipeline evaluates pre-segmented fragments, whereas real video contains continuous speech, so a separate application was built around the same model instances — it imports the pipeline’s model wrapper directly, so the application and the evaluation cannot drift apart. Its stages are listed in Table 3.3.

Stage 5 is the substantive one. Each synthesised segment is stretched or compressed to exactly the duration of the segment it replaces using WORLD analysis and resynthesis (Morise, Yokomori, & Ozawa, 2016), which changes duration without changing pitch, so formants — and therefore the speaker’s identity — survive the operation. Segments are then placed on a silent timeline at their original start times, so the dubbed track has the same length as the source. Scaling is clamped to the range $0.5\times$ to $2.0\times$, beyond which the audio degrades audibly; clamped segments are counted and displayed rather than hidden, and Section 4.9.1 reports what happens when the clamp binds. Stage 6

Table 3.3: Processing stages of the dubbing application. Stage 5 exists because of the measured duration inflation reported in Section 4.9.

Stage	Tool	Operation
1	<code>ffmpeg</code>	Extract the audio track to 16 kHz mono PCM.
2	Whisper	Transcribe with segment timestamps, giving each utterance a slot on the timeline.
3	MarianMT	Translate each segment independently.
4	XTTS-v2	Synthesise each segment, conditioned on the full source audio for voice.
5	WORLD	Time-scale each synthesised segment to the duration of the slot it replaces.
6	<code>ffmpeg</code>	Remux: copy the video stream unchanged, replace the audio.

copies the video stream bit-for-bit, so the picture is never re-encoded.

3.9 Computational Environment and Reproducibility

All experiments were run locally on an Apple silicon machine: the recogniser, synthesiser and speaker encoder on the CPU, which is the stable configuration for this stack on that platform, and translation fine-tuning on the integrated GPU through Metal Performance Shaders. No cloud compute was used and no data left the machine.

Five practices support reproduction. Every experiment is driven by a command-line entry point rather than by a notebook. Every result row is checkpointed to disk as it is produced, so a long run can resume without repeating completed work. Every reported figure and table is regenerated from the committed result files by a script, so no number in Chapter 4 is typed by hand. A provenance record is written beside the analysis outputs, capturing the git commit, the interpreter version, the platform and the versions of every numerical library. And the exact commands that reproduce each result are listed in Appendix A.

3.10 Ethical Considerations

No human participants were recruited and no personal data were collected for this study. The corpus is a public-domain release whose own collection protocol obtained

participant consent (Ward et al., 2023); it is used within its licence and is not redistributed. The pre-trained models are used under their respective licences, and the synthesiser’s non-commercial licence was reviewed and accepted explicitly before use, through a deliberate opt-in rather than a default.

Voice cloning carries a dual-use risk that a dissertation on the subject should state plainly. A system that reproduces a person’s voice in a language they never spoke can be used to fabricate speech attributed to them. Three properties of this work limit that risk without eliminating it: the study operates only on a public-domain corpus whose participants consented to research use; no model was trained to reproduce any specific individual, all cloning being zero-shot from a reference recording; and the application produces no watermark-free artefact beyond what the underlying public model already produces. Any deployment beyond research use should require verified consent from the speaker whose voice is cloned, and should mark generated audio as synthetic.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter reports the four experiments of Section 3.1 in the order they were described, followed by the dubbing application, a methodological finding about the measurement itself, and an overall discussion. All system results are computed on the held-out test split, in both directions, over 870 attempted utterances.

4.1 Corpus Preparation and the Quality Gate

Feature extraction succeeded on all 2,893 linked pairs; no pair was lost to an extraction error. The quality gate of Section 3.5 then removed 589 pairs, and restriction to single-speaker pairs removed a further 49, leaving 2,304 pairs for the corpus study — a drop rate of 20.4 %. Table 4.1 lists what each criterion removed. The criteria overlap, so the individual counts sum to more than the total.

Table 4.1: Pairs removed by each quality criterion, counted if either the English or the Spanish side failed. Criteria overlap, so the counts are not additive. All criteria are properties of the measurement and were applied before any outcome was computed.

Criterion	Pairs affected	Reason
F0 not estimable	229	no voiced frames detected on one side
Voiced-frame ratio < 0.30	524	too little voiced speech to characterise pitch
Duration outside 0.4–20 s	97	fragment too short or too long to measure
Mean F0 outside 60–400 Hz	386	pitch-tracking failure, not a plausible voice
Different linked speakers	49	pair not performed by a single speaker
Retained	2,304	1,634 train / 322 dev / 348 test

Eleven retained pairs have a mean F0 within 5 % of a plausibility bound. They are reported rather than removed, because removing observations by how far they sit from an expected value would make the gate depend on the outcome; the median absolute error is reported alongside the mean throughout Section 4.7 as the robust statistic.

4.2 Pipeline Reliability and Intelligibility

Of 870 attempted utterances, 868 completed and 2 failed, a failure rate of 0.23 %. Both failures occurred at the synthesis stage and share one cause: the reference recording was shorter than the synthesiser’s 0.33 s minimum. The two source utterances are the backchannels “*Mm-hmm.*” and “*Yeah.*” This is a genuine limitation of the pipeline on very short conversational turns, and both rows remain in the results file and in the reported rate.

Table 4.2 gives the recognition and intelligibility results. Recognition is markedly harder on Spanish source audio (corpus WER 0.221) than on English (0.150), which matters later: Section 4.3 shows that this difference propagates into translation quality.

Table 4.2: Recognition and intelligibility on the held-out test split. Corpus figures pool all edits over all reference words; speaker means average within each of the 14 test speakers first, then across speakers, so that no speaker dominates. Lower is better throughout.

Measure	Corpus	Utterance mean	Speaker mean
<i>EN → ES (English source, n = 433)</i>			
Recognition WER	0.150	0.209	0.179
Synthesis intelligibility WER	0.081	0.139	0.135
<i>ES → EN (Spanish source, n = 435)</i>			
Recognition WER	0.221	0.265	0.263
Synthesis intelligibility WER	0.087	0.135	0.114

The synthesis intelligibility figures are low, meaning the generated audio is easy to re-recognise: re-transcribing the output and comparing against the text that was sent to the synthesiser gives a corpus WER of 0.081 and 0.087. Only three of 868 outputs produced an empty re-transcription. The synthesiser is therefore producing clearly articulated speech; whatever it fails to preserve, it is not intelligibility.

The gap between the corpus figure and the utterance mean is worth one remark, because it recurs in every WER row. Corpus WER weights each utterance by its length; the utterance mean weights every utterance equally, so a single wrong word in a three-word backchannel counts as heavily as ten errors in a thirty-word turn. A conversational corpus is full of short turns, so the mean is always the higher figure. Both are reported because they answer different questions, and neither is presented alone.

4.3 Translation Quality and Cascade Error Propagation

Corpus BLEU on the full cascade is 22.14 for $\text{EN} \rightarrow \text{ES}$ and 23.45 for $\text{ES} \rightarrow \text{EN}$, with chrF at 47.22 and 46.63 respectively. These absolute values must be read with the caution of Section 3.4: the reference is a spoken *re-enactment*, so it is a legitimate rendition of the same content but not a sentence-level gloss, and the scores are correspondingly lower than the same models achieve on written parallel text. Differences against the same reference remain interpretable.

The fine-tuning experiment supplies a second measurement of the same models translating the *human* transcript rather than the recogniser output, on the same test utterances and under identical decoding. The difference between the two isolates the cost of the recognition stage (Figure 4.1).

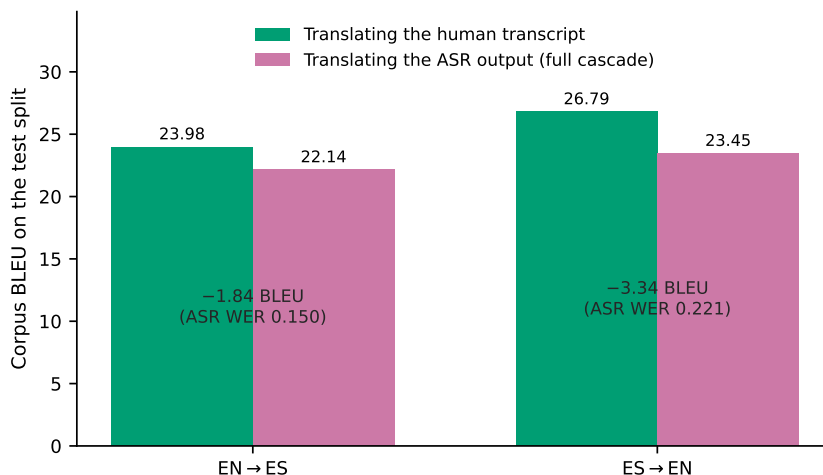


Figure 4.1: Cost of automatic recognition in the cascade. The same translation model, the same test utterances and identical decoding; the only difference is whether the input is the human transcript or the Whisper output. The loss is larger in the direction where recognition is worse.

Recognition errors cost 1.84 BLEU in $\text{EN} \rightarrow \text{ES}$ and 3.34 BLEU in $\text{ES} \rightarrow \text{EN}$. The ordering is the important part: the larger loss occurs in the direction with the higher recognition error rate (0.221 against 0.150). This is the classic weakness of cascaded speech translation (Sperber & Paulik, 2020), measured here on the project’s own data rather than cited, and it is a quantitative argument for the end-to-end approaches reviewed in Section 2.1. It also identifies where effort would pay: on Spanish source audio, an improvement in recognition is worth roughly twice as much downstream as the same improvement on English.

4.4 Speaker Identity Against the Human Ceiling

Speaker similarity between the source recording and the synthesised output is 0.421 for EN $\rightarrow$ ES and 0.383 for ES $\rightarrow$ EN (speaker means). In isolation these numbers are uninterpretable. The same-speaker design supplies the missing scale: the same speaker’s real recording in the other language scores 0.445 in both directions. Against that ceiling the system retains **94.5 %** and **86.1 %** of the achievable similarity (Figure 4.2).

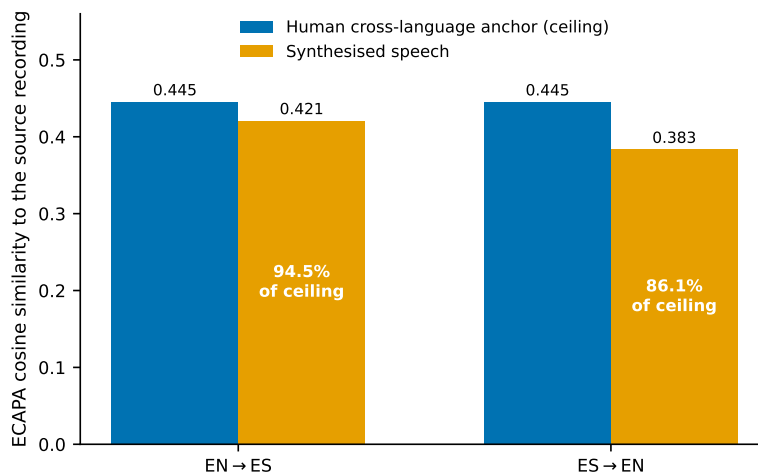


Figure 4.2: Speaker identity measured against the human cross-language ceiling. The ceiling is the ECAPA cosine between the source recording and the *same speaker’s* real recording in the other language; it is available only because the corpus is same-speaker parallel.

Three observations follow.

First, the ceiling itself is low — 0.445, not 0.9. Two genuine recordings of one person, in two languages, are only that similar under an ECAPA cosine. Any evaluation that reports a raw cosine of 0.42 as poor preservation is applying a standard that human speech does not meet either.

Second, the two ceilings are near-identical (0.4452 and 0.4450), as they must be, since both compare the same pair of human recordings with the roles exchanged. The asymmetry between 94.5 % and 86.1 % is therefore entirely in the synthesiser’s output: cloning works better *into* Spanish than into English. A plausible explanation is the language balance of the synthesiser’s training data, but that was not tested here and is offered as a hypothesis, not a finding.

Third, the aggregate conceals real variation. Per speaker, the percentage of ceiling ranges from 74.7 % to 147.2 % in EN $\rightarrow$ ES (median 90.7 %) and from 55.5 % to 107.9 % in ES $\rightarrow$ EN (median 85.8 %). Values above 100 % occur where the synthesised output

resembles the source recording more closely than the speaker’s own other-language recording does — which is possible, because the synthesised audio inherits channel and recording characteristics from the reference clip that the human’s second recording need not share. The test split is also unbalanced, ranging from 2 to 118 pairs per speaker, which is why every aggregate in this chapter is a speaker mean rather than a mean over rows.

4.5 Cross-Lingual Prosody in Human Speech

Before the system can be judged, the human behaviour it is being compared against must be established. Table 4.3 characterises how the same speaker’s prosody differs between their English and Spanish renditions of the same content, over 2,304 pairs and 92 speakers.

Table 4.3: Cross-lingual prosody of the same speaker over 2,304 same-speaker pairs. Δ is the paired mean difference (Spanish minus English) with a 95 % confidence interval. F0 is in semitones relative to 100 Hz.

Measure	English	Spanish	Δ (ES–EN)	95 % CI	Pooled r
F0 level (st)	8.51	8.67	+0.16	[0.08, 0.24]	0.952
F0 range (st)	2.31	2.23	−0.08	[−0.16, −0.00]	0.263
Duration (s)	2.73	2.85	+0.12	[0.09, 0.15]	0.870
Speaking rate (w/s)	3.42	3.07	−0.36	[−0.40, −0.32]	0.595

The systematic language-level differences are small but statistically clear. Spanish renditions sit 0.16 semitones higher in pitch, run 0.12 s longer, and are spoken 0.36 words per second slower; the median Spanish-to-English duration ratio is 1.049 (Figure D.2). All four confidence intervals exclude zero, although Figure D.1 shows that each shift is small relative to the spread of individual pairs. For the purposes of this study, however, the interesting quantity is not the mean shift but the *correspondence*: how strongly one speaker’s English behaviour predicts their own Spanish behaviour. Figure 4.3 shows this for pitch.

Pooled correlations are high for pitch level (0.952) and duration (0.870), moderate for speaking rate (0.595) and weak for pitch range (0.263). The pooled figures are not, however, the right answer, and Section 4.6 decomposes them.

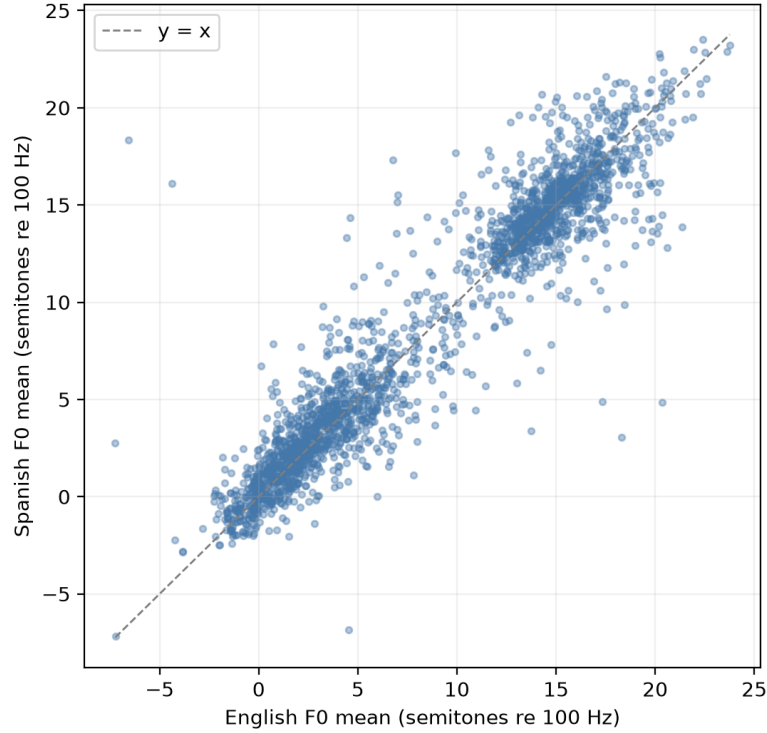


Figure 4.3: English against Spanish mean F0 in semitones for the same speaker and the same content, over 2,304 pairs. The tight relationship along the diagonal is what makes a human cross-language prosody ceiling meaningful.

4.6 The Prosody Transfer Gap

This is the central result of the dissertation. Table 4.4 places the system’s prosody, measured against the human target-language recording, beside the human’s own cross-language correspondence for the same features and the same measurement procedure, and Figure 4.4 plots the within-speaker columns of that table.

Pitch level. A human re-enacting their own utterance in the other language reproduces their pitch level at $r = 0.849$ within-speaker. The system reaches 0.311 and 0.314 — about 37 % of the human figure. The mechanism is known and was stated in advance (Section 2.4): the synthesiser is conditioned on the reference recording for timbre and is never given the source pitch contour, so it must invent intonation from the target text.

Pitch range collapses. Human within-speaker correspondence for F0 range is already modest at 0.214; the system reaches 0.035 and 0.040, which is effectively no relationship at all. Utterance-level expressive variation — the part of prosody that distinguishes an excited turn from a flat one — does not survive the pipeline.

Table 4.4: Cross-lingual prosody correspondence, human against system. Every correlation is with the human target-language recording. *Within-speaker* removes each speaker’s own mean before correlating and is the figure that answers the research question; *pooled* is shown to expose how much of it is speaker identity rather than prosody transfer.

Measure	Human EN $\leftrightarrow$ ES		System EN $\rightarrow$ ES		System ES $\rightarrow$ EN	
	Pooled	Within	Pooled	Within	Pooled	Within
F0 level	0.952	0.849	0.698	0.311	0.754	0.314
F0 range	0.263	0.214	0.142	0.035	0.132	0.040
Duration	0.870	0.867	0.364	0.352	0.618	0.610
Speaking rate	0.595	0.601	0.360	0.372	0.353	0.346
Energy	—	—	0.076	0.090	0.057	0.060
n (pairs)	2,304		345–433		374–435	

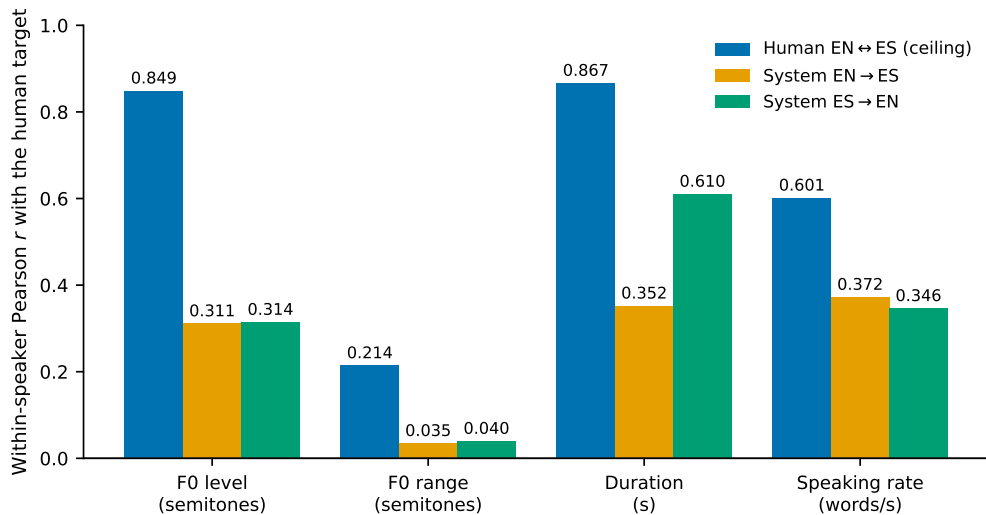


Figure 4.4: Within-speaker cross-lingual prosody correspondence. The blue bars are what a human achieves when re-enacting their own utterance in the other language; the orange and green bars are what the system achieves against that same human recording.

Duration survives best. The system reaches 0.352 and 0.610 against a human 0.867. This is unsurprising: utterance duration is largely determined by how much there is to say, and the translated text preserves the content. It is also the feature the dubbing application can repair by post-processing, which the other features are not.

Energy carries essentially nothing (0.076 and 0.057 pooled), which is expected, since recording level is a property of the microphone and gain rather than of the speaker.

4.6.1 Why the Pooled Figures Must Not Be Quoted Alone

Comparing the pooled and within-speaker columns of Table 4.4 makes the point. For the system’s pitch level, the pooled correlation is 0.698 and 0.754, while the within-speaker correlation is 0.311 and 0.314. The between-speaker correlation — computed over one point per speaker — is 0.895 and 0.847, and speaker differences account for 47 % and 61 % of the total variance in the measurement. Figure 4.5 shows the contrast.

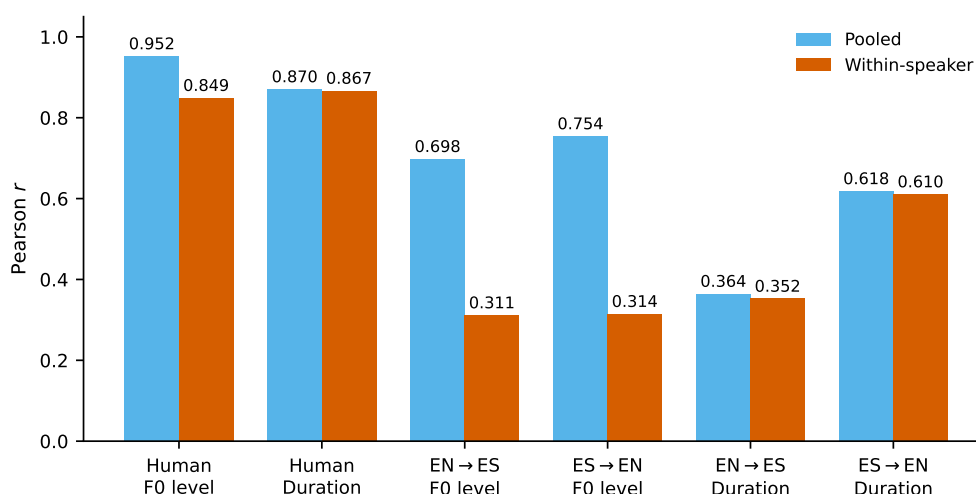


Figure 4.5: Pooled against within-speaker correlation. Where the two differ sharply, as they do for the system’s pitch level, most of the pooled figure is stable between-speaker variation — a low-pitched person remaining low-pitched — carried by voice cloning rather than by prosody transfer. Duration, where the two nearly coincide, is a genuine correspondence.

Read correctly, this is not a weaker version of the result but a sharper one. A pooled correlation near 0.75 would ordinarily be reported as strong prosody transfer. It is not. It is the voice clone reproducing the speaker’s habitual pitch — which is exactly the identity result of Section 4.4 appearing a second time under a different name. Once speaker identity is removed, what remains is 0.31: the same pipeline that preserves identity well preserves utterance-level prosody poorly. Reporting the pooled figure alone would have attributed the success of one component to another.

4.6.2 Relation to the Literature

This quantifies precisely the gap that expressive dubbing systems exist to close. Swiatkowski et al. (2023) developed a cross-lingual prosody representation for machine dubbing without access to parallel expressive data; Song et al. (2023) pursued the same goal inside a direct S2ST model. The contribution here is complementary rather than competing: it is not a better system but a calibrated statement of the size of the problem, expressed in units — a within-speaker correlation against a human ceiling — that any subsequent system can be measured in.

4.7 Predicting Target-Language Prosody

Section 4.6 shows the system does not transfer prosody. This section asks a narrower question: is target-language prosody *predictable* from the source at all? If it is not, no amount of conditioning would help.

Table 4.5 reports the test-split result for each direction and target, against the three baselines of Section 3.6. Figure 4.6 shows the same comparison graphically.

Table 4.5: Prosody prediction on the held-out test split, in mean absolute error (lower is better). Δ is the change relative to the copy-plus-offset baseline B2; p is a Wilcoxon signed-rank test on per-speaker mean absolute errors ($n = 14$ speakers), which is the appropriate unit of independence. F0 targets are in semitones, duration in log-ratio.

Direction	Target	B1	B0	B2	Ridge	Δ vs B2	p
EN $\rightarrow$ ES	F0 level	5.783	1.375	1.353	1.353	−0.02 %	0.0040
EN $\rightarrow$ ES	F0 range	1.034	0.977	0.983	0.932	−5.14 %	0.0012
EN $\rightarrow$ ES	Duration ratio	0.181	0.186	0.181	0.180	−0.73 %	0.1937
ES $\rightarrow$ EN	F0 level	5.947	1.350	1.329	1.329	−0.00 %	0.9032
ES $\rightarrow$ EN	F0 range	1.017	0.963	0.968	0.931	−3.89 %	0.0001
ES $\rightarrow$ EN	Duration ratio	0.182	0.187	0.182	0.172	−5.34 %	0.1040

4.7.1 What the Result Says

Pitch level is not predictable beyond the baseline. The improvement is −0.02 % and −0.00 %: numerically nothing. The regulariser selected $\lambda = 10^6$ in both directions, at which every weight is effectively zero, so the model *chose* to become the baseline. This

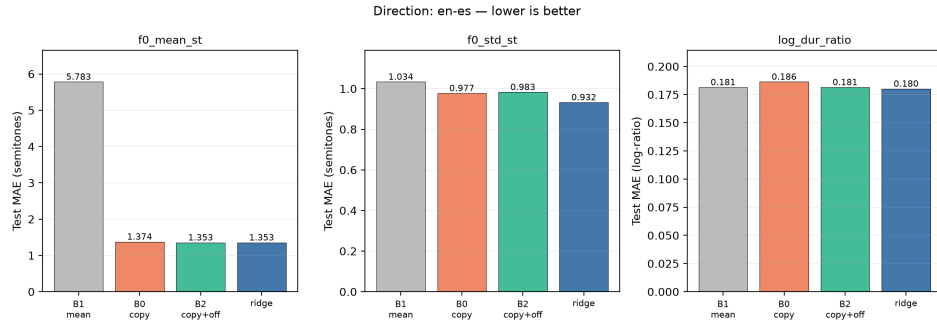


Figure 4.6: Test mean absolute error for each prosody target in the EN $\rightarrow$ ES direction, model against the three baselines. The trivial baseline B1, which ignores the source entirely, is catastrophic for pitch level; the informative comparison is against B2.

is not a failure of fitting but the regulariser reporting honestly that no conditional signal is available. It is also strong evidence against overfitting.

Pitch range is predictable, in both directions. The model reduces mean absolute error by 5.14 % (EN $\rightarrow$ ES) and 3.89 % (ES $\rightarrow$ EN), with speaker-clustered p of 0.0012 and 0.0001 and bootstrap confidence intervals for the difference that exclude zero; Figure D.3 in Appendix D shows the ES $\rightarrow$ EN direction. The mechanism is visible in the fitted coefficients (Table B.2): the weight on source F0 range is -0.191 and -0.143 — large relative to the other five weights, and *negative*. The model has learned that an unusually wide source pitch range should be predicted narrower in the target: per-utterance variation regresses towards the speaker’s typical range rather than transferring one for one. That is an interpretable finding about cross-lingual prosody, not merely a number.

Duration ratio is not reliably predictable. The nominal improvement of 5.34 % in ES $\rightarrow$ EN does not survive clustering by speaker ($p = 0.104$).

4.7.2 Multiple Comparisons

Six models were fitted, so the family-wise position must be stated rather than left implicit. Under Bonferroni at $\alpha = 0.05$ the threshold is $0.05/6 = 0.0083$; three of the six speaker-clustered p -values fall below it (0.0001, 0.0012, 0.0040), and the same three survive Holm and Benjamini–Hochberg at 5 %. Only two of those three are practically meaningful: the F0 range results in both directions. The third, EN $\rightarrow$ ES pitch level, is statistically detectable and practically nil, an effect of -0.02% that arises because the model is numerically almost identical to the baseline it is being compared against — a signed-rank test over near-identical error vectors can return a small p for a zero-sized

effect. It is reported here so that it cannot be quoted as a third success.

The defensible claim is therefore narrow and consistent: *the predictable component of cross-lingual prosody lies in pitch range, not in pitch level or timing*, supported by agreement in sign and magnitude across both directions, by survival of family-wise correction, and by a coefficient whose value explains the mechanism.

4.8 Fine-Tuning the Translation Component

Fine-tuning on 1,998 in-domain sentence pairs improved reference-matching in both directions, on a held-out test set of unseen speakers under identical decoding (Figure 4.7). For $\text{EN} \rightarrow \text{ES}$, BLEU rose from 23.98 to 26.31 and chrF from 49.87 to 51.70, a gain of +2.34 and +1.82. For $\text{ES} \rightarrow \text{EN}$, BLEU rose from 26.79 to 30.02 and chrF from 50.30 to 52.27, a gain of +3.23 and +1.97.

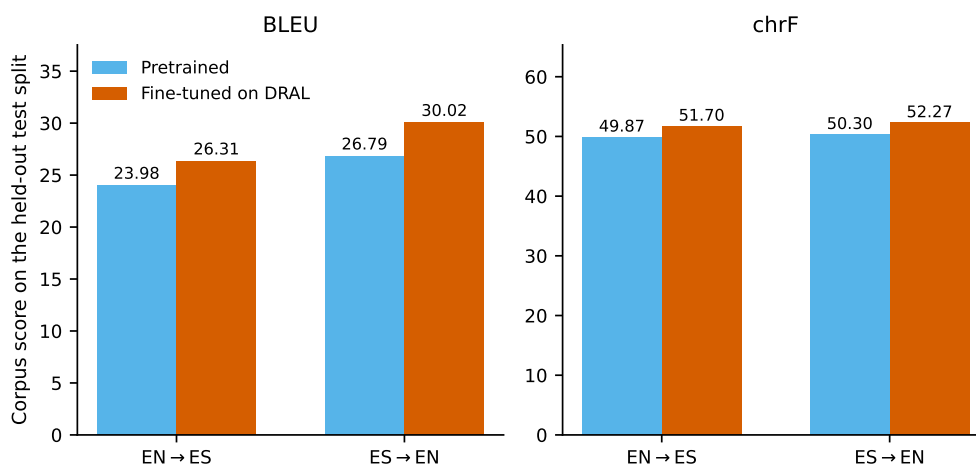


Figure 4.7: Translation quality before and after in-domain fine-tuning, on the held-out test split ($n = 432$ per direction) with identical decoding settings for both systems.

The checkpoint was selected on development loss, and the selection is the methodologically important part. Table 4.6 shows textbook overfitting: training loss falls monotonically while development loss reaches its minimum at epoch 2 and rises thereafter. Both directions independently selected epoch 2, as Figures D.4 and D.5 in Appendix D show. The test split was scored once, after selection.

Table 4.6: Training and development loss per epoch. Development loss is minimised at epoch 2 in both directions, and the epoch-2 checkpoint was selected on that basis before the test split was touched.

Epoch	EN $\rightarrow$ ES		ES $\rightarrow$ EN	
	Train	Dev	Train	Dev
1	1.922	1.730	1.614	1.378
2	1.569	1.712	1.337	1.361
3	1.368	1.744	1.165	1.382
4	1.221	1.790	1.032	1.432

The BLEU gain is register and style adaptation, not improved translation adequacy. The corpus’s Spanish side is a re-enactment, so matching the reference more closely means sounding more like these particular speakers, not translating better. In the fine-tuned outputs, bicicleta becomes bici and presa becomes represa; nos ayudamos mutuamente becomes nos ayudamos el uno al otro. In several cases the pre-trained output is the better standalone translation. That chrF moves less than BLEU (+1.8 against +2.3) is consistent with this reading, since chrF is character-based and comparatively robust to lexical substitution and word-order change.

4.8.1 Why Only the Translation Component Was Adapted

Recognition already achieves a corpus WER of 0.150 on English source audio, leaving little headroom, and adapting a large recogniser on 1,998 utterances of one corpus would risk harming generalisation. Adapting the synthesiser requires substantially more data and compute than a laptop provides. Translation is the cheapest experiment with a clear domain mismatch to exploit and a well-understood metric: 12.4 and 10.8 minutes of training respectively. The choice is an engineering judgement about where a limited budget buys the most information, and it is stated as such.

4.8.2 Reconciling the Progress-Report BLEU

The progress report submitted in January 2026 quoted a BLEU of approximately 61, whereas this study measures the pre-trained model at 23.98. Both figures are correct, and they measure different things. The earlier figure was a *back-translation* score: the Spanish output was translated back to English and compared against the original English

text. That is a much easier comparison, because errors introduced on the outward pass can cancel on the return. The figures in this chapter are computed against an independent human reference. The change is a change of instrument, not a regression in the system.

4.9 Duration Inflation and the Retiming Stage

Raw synthesised speech is systematically longer than the speech it replaces. Averaged over the test split, generated utterances run 4.07 s against a human target of 2.67 s, a ratio of $1.52\times$; the median per-utterance ratio is 1.43, and 25.1 % of utterances exceed $2.0\times$ (Figure 4.8). The inflation is worse in EN $\rightarrow$ ES (median 1.51) than in ES $\rightarrow$ EN (median 1.32).

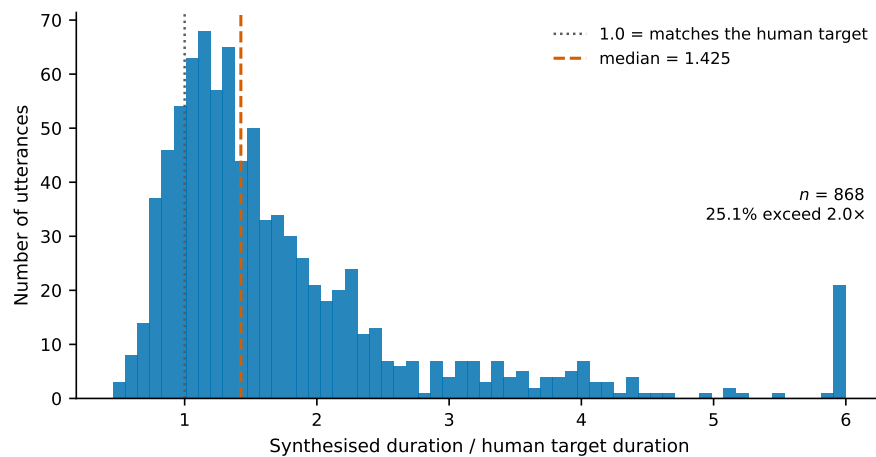


Figure 4.8: Ratio of synthesised duration to human target duration over all 868 successful utterances, clipped at $6\times$ for display. The distribution sits clearly to the right of 1.0: the inflation is a systematic bias, not random error.

This is not a subtle effect and it is fatal for dubbing: at $1.5\times$ per utterance, a dubbed track drifts out of synchronisation with the picture within a few turns. It is the reason the application includes the WORLD retiming stage described in Section 3.8.

4.9.1 Behaviour of the Dubbing Application

Table 4.7 reports five real dubbing jobs, 83 segments in total. The retiming stage works, and its limit is visible.

Where the raw ratio is at or below about $1.6\times$, retiming brings the dubbed track to within 7 % of the original length and the clamp binds on at most a quarter of segments. Where

Table 4.7: Five dubbing jobs. *Raw ratio* is synthesised duration over the original slot before retiming; *final match* is the dubbed track length over the original after retiming, where 1.000 is perfect. *Clamped* counts segments needing a scale outside the permitted $0.5\times$ – $2.0\times$ range.

Job	Direction	Segments	Clamped	Raw ratio	Final match	Identity
f44255c7	ES $\rightarrow$ EN	3	0 (0 %)	0.785	1.001	0.422
dafbca94	ES $\rightarrow$ EN	4	1 (25 %)	1.243	1.056	0.718
0d472aee	ES $\rightarrow$ EN	22	5 (23 %)	1.510	1.062	0.619
44e83311	EN $\rightarrow$ ES	15	1 (7 %)	1.583	1.067	0.385
3ae892f4	EN $\rightarrow$ ES	39	23 (59 %)	2.173	1.226	0.260

the raw ratio reaches $2.17\times$, the clamp binds on 59 % of segments, and the final track is 22.6 % too long. The failure mode is worth stating precisely, because it explains what a listener hears: a 0.84 s slot receiving 7.61 s of synthesised speech would need a scale of $0.11\times$; the floor of $0.5\times$ produces 3.81 s, which is 2.97 s too long, and that overflow lands on top of the following segment. Segments are summed into the timeline, so two voices speak at once.

The diagnosis identifies the correct fix, and it is upstream. No amount of digital signal processing can absorb a $2.17\times$ length ratio without destroying the audio. The real problem is that the translation model produces an utterance whose spoken length was never constrained — exactly the problem that length-controlled translation for dubbing was designed to solve (Lakew et al., 2021; Federico et al., 2020). This is the first item of future work in Section 5.5.

Two secondary observations. Speaker identity measured on the retimed output ranges from 0.260 to 0.718 across jobs, and the lowest value is the job with the highest clamp rate, which is consistent with heavy time-scaling degrading the voice. And the pipeline is slow: roughly $13\times$ real time on the test machine, because synthesis is inherently sequential and runs on the CPU.

4.10 A Correction to the Measurement Itself

One finding of this project is about its own instrument, and it is reported because the earlier conclusion it overturned was wrong by roughly a factor of seven.

An earlier version of the analysis reported that the system transferred essentially no pitch information: a correlation of 0.090 between generated and human-target F0. Two

defects produced that number.

Correlating pitch in hertz. Pitch perception is logarithmic, so a correlation computed in hertz is dominated by whichever values lie furthest from the mean, and a single octave-doubled observation is enough to destroy it. On identical underlying measurements, correlating in semitones instead of hertz moved the EN → ES F0 correlation from 0.048 to 0.368.

No plausibility gate. The pitch tracker’s search band admits occasional octave doublings. Excluding observations whose median F0 falls outside the plausible adult speaking range — a criterion on the measurement, not on the outcome — moved the same figure from 0.368 to 0.820.

A third intervention, attempted first, made matters worse and is recorded because the reasoning behind it was wrong in an instructive way. Narrowing the pitch tracker’s *search* band to 60–400 Hz seemed the natural remedy for octave errors. Measured, it was the worst of the four bands tried:

Search band	EN → ES	ES → EN
60–400 Hz	0.371	0.589
65–600 Hz	0.778	0.739
65–1000 Hz	0.888	0.856
65–2093 Hz	0.822	0.819

The cause is that the tracker’s voiced/unvoiced decision depends on how many frequency candidates it has: narrowing the band lowers the aggregate voiced probability until, on quiet recordings, it detects no voicing at all. One test file went from ten voiced frames to zero. The correct design separates the two bands — a wide search band so voicing detection works, and a narrow plausibility band applied afterwards as a data-quality gate — which is the procedure documented in Section 3.4.5 and used throughout this chapter.

Three lessons are worth recording. A measurement artefact can masquerade as a strong scientific finding, and “the system transfers no prosody” was a more publishable claim than the truth. The plausible fix was the wrong fix, and only measuring the alternatives revealed it. And the corrected result is still a negative one — within-speaker transfer of 0.31 against a human 0.85 — but it is now a defensible negative rather than an artefact.

4.11 Discussion

Taken together, the results support one clear statement: **zero-shot cross-lingual voice cloning preserves speaker identity well and utterance-level prosody poorly, and the two should not be reported under a single heading.**

The identity result is genuinely good, and it is only recognisable as good because of the ceiling. Reported as a bare cosine of 0.42, the system looks mediocre; measured against the 0.445 that two real recordings of the same person achieve, it retains 86–95 % of what is available. The prosody result is genuinely poor, and it is only recognisable as poor because the same instrument supplies the human comparison: 0.31 against 0.85 within-speaker for pitch level, and effectively zero against 0.21 for pitch range.

That the two results diverge so sharply is explained by the architecture, not by an implementation defect. The speaker embedding on which cloning depends is trained for verification and is designed to be invariant to what a speaker does on any particular utterance — which is precisely the information prosody consists of. A system built on such an embedding will reproduce who is speaking and discard how they spoke. Closing that gap requires an additional conditioning path, which is what the expressive dubbing literature builds (Swiatkowski et al., 2023; Song et al., 2023).

The prediction experiment adds a constructive qualification. Cross-lingual prosody is not arbitrary, but the part of it that is predictable from the source is narrower than one might hope: pitch range, not pitch level or timing, and the relationship is compressive rather than one-for-one. A conditioning mechanism that copied source pitch directly into the target would be copying the wrong thing.

Two further findings are practically useful. The cascade loses 1.8–3.3 BLEU to recognition errors, in proportion to the error rate, which locates the weakest link. And synthesised speech runs $1.5\times$ too long, which makes duration control — not audio post-processing — the binding constraint on usable dubbing.

4.12 Limitations

1. **No listening study.** Every result is instrumental. Whether listeners perceive the prosody gap as strongly as the correlations imply is untested. This is the most

significant gap and the first item of future work.

2. **One corpus, one language pair, fourteen test speakers.** The test split is small in speakers even though it is large in utterances, which is why speaker-clustered inference is used throughout. The findings are stated for English–Spanish conversational re-enactment and are not claimed to generalise to other language pairs or to read speech.
3. **Re-enactment references.** The corpus’s target side is a performance of the same communicative content, not a literal translation. BLEU and chrF against it are therefore conservative and measure a different thing from adequacy on written parallel text.
4. **Prosody is measured with utterance-level statistics.** Mean and standard deviation of F0 do not capture contour shape. A frame-level comparison under dynamic time warping would be a stronger instrument and would make these results directly comparable with published expressive-dubbing scores.
5. **PESQ and STOI are not reported,** for the reason given in Section 3.4.6: they require a time-aligned reference of the same utterance, which cannot exist across languages.
6. **The prosody model is analysis, not control.** It predicts target-language prosody; it does not condition the synthesiser, so it does not improve the audio.
7. **Effect sizes are small where they are significant.** The pitch-range improvement is 3.9–5.1 % of mean absolute error. It is real, replicated across directions and family-wise significant, but it is not large.
8. **The dubbing application is single-speaker and speech-only.** It does not diarise multiple speakers, preserve background music, or perform lip synchronisation.

CHAPTER 5

CONCLUSIONS

5.1 Overall Assessment

This study set out to build a bilingual English–Spanish voice transcription and synthesis system that preserves the speaker’s identity, and to establish how much of the speaker’s prosody it carries. Both were achieved, and the second produced a result that is more informative than the first.

The system works and is reliable: 868 of 870 held-out utterances completed, the two failures having a single identified cause, and the synthesised speech is easy to re-recognise (corpus intelligibility WER 0.081 and 0.087). Speaker identity is preserved well — 94.5 % and 86.1 % of the human cross-language ceiling. Prosody is not: within a speaker, pitch level transfers at 0.31 against a human 0.85, and pitch range essentially not at all.

The strength of the study lies less in the system than in what the corpus permitted. Because one bilingual speaker performs both sides of every pair, each measurement has a human answer beside it. That turns two uninterpretable numbers into calibrated ones and separates two properties that are routinely conflated: a pipeline can preserve *who is speaking* very well while preserving *how they spoke* very poorly, and only a same-speaker reference makes that visible.

5.2 Conclusions Against the Objectives

Objective 1 — bidirectional voice-preserving translation framework: achieved.

A cascade of Whisper, MarianMT and XTTS-v2 was implemented and evaluated in both directions on a speaker-disjoint split, with deterministic per-utterance seeding and every failure retained in the results. Identity preservation was measured against a human

cross-language anchor rather than reported as an unanchored cosine, and reaches 94.5 % (EN → ES) and 86.1 % (ES → EN) of that ceiling. The asymmetry between directions is a property of the synthesiser, since the two ceilings are identical to three decimal places.

Objective 2 — prosody analysis, transfer measurement and prediction: achieved, with a negative result. The human cross-lingual prosodic relationship was characterised over 2,304 same-speaker pairs, decomposed into within- and between-speaker components, and used as the reference for the system. The transfer gap was quantified feature by feature. A supervised mapping from source to target prosody improved on a copy-plus-offset baseline only for pitch range, by 5.1 % and 3.9 % of mean absolute error, surviving family-wise correction in both directions; pitch level and timing were not predictable beyond the baseline. The learned coefficient on source pitch range is negative, which supplies the mechanism: utterance-level variation regresses towards the speaker’s typical range rather than transferring one for one.

Objective 3 — integration with video: achieved, with a stated limit. A web application extracts audio, transcribes with timestamps, translates and synthesises each segment in the original speaker’s voice, retimes each segment to its slot with WORLD analysis and resynthesis, and remuxes without re-encoding the picture. On five real jobs the dubbed track lands within 7 % of the original length when the raw length ratio is at or below about $1.6\times$. Where it reaches $2.17\times$, the scaling clamp binds on 59 % of segments and the result degrades. The cause was diagnosed as unconstrained translation length rather than a fault in the retiming.

Additional activity — in-domain fine-tuning: completed, and correctly qualified. Fine-tuning raised BLEU by 2.34 and 3.23 with proper development-set checkpoint selection and a single scored pass over the test split. The gain is register adaptation to the corpus’s conversational style, not improved translation adequacy, and is reported as such.

5.3 Answers to the Research Questions

RQ1 86–95 % of the achievable speaker identity survives. Measured against the same speaker’s real recording in the other language, the system retains 94.5 % of the ceiling into Spanish and 86.1 % into English. The ceiling itself is only 0.445 under an ECAPA cosine, which is the fact that makes the raw score interpretable.

RQ2 Humans transfer prosody strongly; the system transfers a fraction of it.

Within-speaker, human cross-language correspondence is 0.849 for pitch level, 0.867 for duration, 0.601 for speaking rate and 0.214 for pitch range. The system reaches 0.311/0.314, 0.352/0.610, 0.372/0.346 and 0.035/0.040 respectively. Pooled correlations are much higher but are largely between-speaker variance carried by voice cloning, and must not be quoted alone.

- RQ3 Only partly.** Target-language pitch range is predictable from source-language prosody beyond a copy-plus-offset baseline, in both directions and under family-wise correction, but the effect is small (3.9–5.1 % of mean absolute error). Pitch level and duration ratio are not. For pitch level the regulariser selected $\lambda = 10^6$ and the model reduced itself to the baseline, which is the correct response to an absent signal.
- RQ4 Recognition errors cost 1.84 and 3.34 BLEU**, in proportion to the recognition error rate in each direction. In-domain fine-tuning of the translation model recovers a comparable magnitude (+2.34 and +3.23 BLEU), but it does so by matching the corpus’s register rather than by translating better, so the two effects should not be netted against each other.

5.4 Contributions

1. **A calibrated identity measurement.** Speaker similarity expressed as a percentage of a human same-speaker cross-language ceiling, rather than as an unanchored cosine.
2. **A quantified prosody transfer gap**, reported within-speaker so that the identity result is not counted twice, with a demonstration that the pooled figure overstates transfer by roughly a factor of two for pitch level.
3. **A negative result with a mechanism.** Cross-lingual prosody is predictable in pitch range and not in pitch level or timing, and the sign of the learned coefficient explains why.
4. **A measured cost for the cascade**, 1.8–3.3 BLEU, obtained from the project’s own data rather than cited.
5. **A methodological correction** showing that correlating pitch in hertz without a plausibility gate understated F0 correspondence by roughly a factor of seven, together with evidence that the intuitive remedy — narrowing the pitch tracker’s search band — makes the measurement worse.
6. **A working, reproducible artefact:** a command-line research pipeline in which every reported number regenerates from committed result files, and a dubbing

application that shares its model instances so that the two cannot drift apart.

5.5 Recommendations and Future Work

The following are ordered by the ratio of value to effort, as judged from the results above.

1. **Conduct a listening study.** Twenty clips spanning good and poor cases, five to ten bilingual listeners, rating naturalness, speaker similarity and prosodic appropriateness on a five-point scale, with the automatic speaker-similarity scores correlated against the human similarity ratings. This validates the instrument as well as the system and closes the most significant gap in the present study.
2. **Train a length-controlled translation model.** Prepend a length-class token to each source sentence during fine-tuning, derived from the target-to-source character ratio, and request a short rendition when the available slot is tight. This is the same twelve-minute fine-tune already performed, it attacks the $1.5\times$ duration inflation at its source rather than in post-processing, and it is measurable in four ways: BLEU, length-compliance rate, clamp rate, and final duration match.
3. **Repair the two application defects identified in Section 4.9.1:** truncate an overflowing segment at the next segment’s start instead of summing overlapping audio, and fill stretches where no speech was detected with the ducked original audio instead of digital silence.
4. **Compare F0 contours frame by frame under dynamic time warping.** Utterance-level mean and standard deviation cannot capture contour shape; a warped frame-level comparison would be a stronger instrument and would make these results directly comparable with published expressive-dubbing scores.
5. **Close the loop on the prosody model.** The pitch-range relationship is real but currently only descriptive. Using the prediction to condition synthesis — and measuring whether within-speaker correspondence improves — would convert an analysis into an intervention.
6. **Merge short recogniser segments before translation.** Sub-second slots produce impossible scaling demands in the dubbing application; merging adjacent short segments would remove a large share of the clamped cases at low cost.
7. **Extend beyond one corpus and one language pair.** Replicating the ceiling methodology on a second same-speaker parallel corpus, or a typologically different language pair, would establish how far these figures generalise.

5.6 Closing Statement

The most durable outcome of this project is not the pipeline, which is assembled from openly available components, but the measurement it makes possible. A same-speaker parallel corpus turns speech synthesis evaluation from a set of uncalibrated similarity scores into a comparison against what the same person actually did. Under that comparison, the answer is specific: current zero-shot cross-lingual synthesis carries the speaker, not the speech act. Identity survives at 86–95 % of what is achievable; utterance-level prosody survives at roughly a third for pitch level and not at all for pitch range. Stating that gap in reusable units is what this dissertation contributes, and it is the quantity that the next generation of expressive dubbing systems will have to move.

REFERENCES

- Ambadekar, S., Dhokiya, T., Gandhi, H., & Haria, I. (2025). A review on multilingual voice-to-voice translation. *Journal of All India Institute of Speech and Hearing*, 44(1), 1–23.
- Ardila, R., Branson, M., Davis, K., Kohler, M., Meyer, J., Henretty, M., . . . Weber, G. (2020). Common voice: A massively-multilingual speech corpus. In *Proceedings of the 12th language resources and evaluation conference (lrec)* (pp. 4218–4222).
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*, 57(1), 289–300.
- Bentivogli, L., Cettolo, M., Gaido, M., Karakanta, A., Martinelli, A., Negri, M., & Turchi, M. (2021). Cascade versus direct speech translation: Do the differences still make a difference? In *Proceedings of the 59th annual meeting of the association for computational linguistics (acl)* (pp. 2873–2887).
- Casanova, E., Davis, K., Gölge, E., Gökmar, G., Gulea, I., Hart, L., . . . Gascó, J. (2024). XTTS: A massively multilingual zero-shot text-to-speech model. In *Proceedings of interspeech* (pp. 4978–4982).
- Casanova, E., Weber, J., Shulby, C., Candido Junior, A., Gölge, E., & Ponti, M. A. (2022). YourTTS: Towards zero-shot multi-speaker TTS and zero-shot voice conversion for everyone. In *Proceedings of the 39th international conference on machine learning (icml)* (pp. 2709–2720). PMLR.
- Desplanques, B., Thienpondt, J., & Demuynck, K. (2020). ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification. In *Proceedings of interspeech* (pp. 3830–3834).
- Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. New York: Chapman and Hall/CRC.
- Federico, M., Enyedi, R., Barra-Chicote, R., Giri, R., Isik, U., Krishnaswamy, A., & Sawaf, H. (2020). From speech-to-speech translation to automatic dubbing. In *Proceedings of the 17th international conference on spoken language translation (iwslt)* (pp. 257–264).
- Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 55–67.
- Inaguma, H., Popuri, S., Kulikov, I., Chen, P.-J., Wang, C., Chung, Y.-A., . . . Pino, J. (2023). UnitY: Two-pass direct speech-to-speech translation with discrete units. In *Proceedings of the 61st annual meeting of the association for computational linguistics (acl)* (pp. 15655–15680).
- Jia, Y., Ramanovich, M. T., Remez, T., & Pomerantz, R. (2022). Translatotron 2: High-quality direct speech-to-speech translation with voice preservation. In *Proceedings of the 39th international conference on machine learning (icml)* (pp. 10120–10134). PMLR.
- Jia, Y., Ramanovich, M. T., Wang, Q., & Zen, H. (2022). CVSS corpus and massively multilingual speech-to-speech translation. In *Proceedings of the 13th language resources and evaluation conference (lrec)* (pp. 6691–6703).
- Jia, Y., Zhang, Y., Weiss, R. J., Wang, Q., Shen, J., Ren, F., . . . Wu, Y. (2018). Transfer

- learning from speaker verification to multispeaker text-to-speech synthesis. In *Advances in neural information processing systems (neurips)* (Vol. 31, pp. 4480–4490).
- Junczys-Dowmunt, M., Grundkiewicz, R., Dwojak, T., Hoang, H., Heafield, K., Necker-mann, T., . . . Birch, A. (2018). Marian: Fast neural machine translation in C++. In *Proceedings of acl 2018, system demonstrations* (pp. 116–121).
- Kim, J., Kong, J., & Son, J. (2021). Conditional variational autoencoder with adversarial learning for end-to-end text-to-speech. In *Proceedings of the 38th international conference on machine learning (icml)* (pp. 5530–5540). PMLR.
- Kong, J., Kim, J., & Bae, J. (2020). HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis. In *Advances in neural information processing systems (neurips)* (Vol. 33, pp. 17022–17033).
- Lakew, S. M., Federico, M., Wang, Y., Hoang, C., Virkar, Y., Barra-Chicote, R., & Enyedi, R. (2021). Machine translation verbosity control for automatic dubbing. In *Proceedings of the ieee international conference on acoustics, speech and signal processing (icassp)* (pp. 7538–7542).
- Lee, A., Chen, P.-J., Wang, C., Gu, J., Popuri, S., Ma, X., . . . Hsu, W.-N. (2022). Direct speech-to-speech translation with discrete units. In *Proceedings of the 60th annual meeting of the association for computational linguistics (acl)* (pp. 3327–3339).
- Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. In *Proceedings of the 7th international conference on learning representations (iclr)*.
- Mauch, M., & Dixon, S. (2014). pYIN: A fundamental frequency estimator using probabilistic threshold distributions. In *Proceedings of the ieee international conference on acoustics, speech and signal processing (icassp)* (pp. 659–663).
- McFee, B., Raffel, C., Liang, D., Ellis, D. P. W., McVicar, M., Battenberg, E., & Nieto, O. (2015). librosa: Audio and music signal analysis in Python. In *Proceedings of the 14th python in science conference (scipy)* (pp. 18–24).
- Morise, M., Yokomori, F., & Ozawa, K. (2016). WORLD: A vocoder-based high-quality speech synthesis system for real-time applications. *IEICE Transactions on Information and Systems*, E99-D(7), 1877–1884.
- Nagrani, A., Chung, J. S., & Zisserman, A. (2017). VoxCeleb: A large-scale speaker identification dataset. In *Proceedings of interspeech* (pp. 2616–2620).
- Nuthakki, P., Katamaneni, M., J. N., C. S., Gubbala, K., Domathoti, B., Maddumala, V. R., & Jetti, K. R. (2023). Deep learning based multilingual speech synthesis using multi feature fusion methods. *ACM Transactions on Asian and Low-Resource Language Information Processing*.
- Papineni, K., Roukos, S., Ward, T., & Zhu, W.-J. (2002). BLEU: A method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the association for computational linguistics (acl)* (pp. 311–318).
- Peng, Y., Kulikov, I., Yang, Y., Popuri, S., Lu, H., Wang, C., & Gong, H. (2024). MSLM-S2ST: A multitask speech language model for textless speech-to-speech translation with speaker style preservation. *arXiv preprint arXiv:2403.12408*.
- Popović, M. (2015). chrF: Character n-gram f-score for automatic MT evaluation. In *Proceedings of the tenth workshop on statistical machine translation (wmt)* (pp. 392–395).
- Post, M. (2018). A call for clarity in reporting BLEU scores. In *Proceedings of the third conference on machine translation (wmt)* (pp. 186–191).
- Qian, K., Zhang, Y., Chang, S., Yang, X., & Hasegawa-Johnson, M. (2019). AutoVC:

- Zero-shot voice style transfer with only autoencoder loss. In *Proceedings of the 36th international conference on machine learning (icml)* (pp. 5210–5219). PMLR.
- Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2023). Robust speech recognition via large-scale weak supervision. In *Proceedings of the 40th international conference on machine learning (icml)* (pp. 28492–28518). PMLR.
- Ravanelli, M., Parcollet, T., Plantinga, P., Rouhe, A., Cornell, S., Lugosch, L., ... Bengio, Y. (2021). SpeechBrain: A general-purpose speech toolkit. *arXiv preprint arXiv:2106.04624*.
- Seamless Communication, Barrault, L., Chung, Y.-A., Coria Meglioli, M., Dale, D., Dong, N., ... Ye, E. (2023). Seamless: Multilingual expressive and streaming speech translation. *arXiv preprint arXiv:2312.05187*.
- Shen, J., Pang, R., Weiss, R. J., Schuster, M., Jaitly, N., Yang, Z., ... Wu, Y. (2018). Natural TTS synthesis by conditioning WaveNet on mel spectrogram predictions. In *Proceedings of the IEEE international conference on acoustics, speech and signal processing (icassp)* (pp. 4779–4783).
- Skerry-Ryan, R. J., Battenberg, E., Xiao, Y., Wang, Y., Stanton, D., Shor, J., ... Saurous, R. A. (2018). Towards end-to-end prosody transfer for expressive speech synthesis with Tacotron. In *Proceedings of the 35th international conference on machine learning (icml)* (pp. 4693–4702). PMLR.
- Song, K., Ren, Y., Lei, Y., Wang, C., Wei, K., Xie, L., ... Ma, Z. (2023). StyleS2ST: Zero-shot style transfer for direct speech-to-speech translation. *arXiv preprint arXiv:2305.17732*.
- Sperber, M., & Paulik, M. (2020). Speech translation and the end-to-end promise: Taking stock of where we are. In *Proceedings of the 58th annual meeting of the association for computational linguistics (acl)* (pp. 7409–7421).
- Swiatkowski, J., Wang, D., Babianski, M., Lumban Tobing, P., Vipplerla, R., & Pollet, V. (2023). Cross-lingual prosody transfer for expressive machine dubbing. In *Proceedings of interspeech* (pp. 4838–4842).
- Tiedemann, J., & Thottingal, S. (2020). OPUS-MT — building open translation services for the world. In *Proceedings of the 22nd annual conference of the European association for machine translation (eamt)* (pp. 479–480).
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... Polosukhin, I. (2017). Attention is all you need. In *Advances in neural information processing systems (neurips)* (Vol. 30, pp. 5998–6008).
- Wang, Y., Stanton, D., Zhang, Y., Skerry-Ryan, R. J., Battenberg, E., Shor, J., ... Saurous, R. A. (2018). Style tokens: Unsupervised style modeling, control and transfer in end-to-end speech synthesis. In *Proceedings of the 35th international conference on machine learning (icml)* (pp. 5180–5189). PMLR.
- Ward, N. G., Avila, J. E., Rivas, E., & Marco, D. (2023). *Dialogs re-enacted across languages, version 2* (Tech. Rep. No. UTEP-CS-23-27). El Paso, TX, USA: Department of Computer Science, University of Texas at El Paso.
- Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6), 80–83.
- Yang, Z. (2025). *Deep emotional text-to-speech and voice conversion* (Unpublished doctoral dissertation). Universität Augsburg.

APPENDIX A

Reproducing Every Result

The commands below regenerate every number, table and figure in Chapter 4 from the raw corpus. They are listed in dependency order. Long-running steps are marked; the remainder run in seconds.

1. Environment.

```
brew install python@3.11 ffmpeg sox git-lfs
python3.11 -m venv .venv && source .venv/bin/activate
python -m pip install -e '[ml,dev,web]'
export COQUI_TOS_AGREED=1      # after reviewing the Coqui model licence
```

2. Corpus, manifest and speaker-disjoint split.

```
tar -xzf data/raw/DRAL-16kHz.tgz -C data/raw
bvt inspect-dral data/raw
bvt build-manifest data/raw data/processed/dral_manifest.csv
bvt split-manifest data/processed/dral_manifest.csv \
    data/splits/dral_manifest.csv --seed 498
bvt merge-transcripts data/splits/dral_manifest.csv \
    data/splits/dral_manifest_with_text.csv \
    --transcripts data/raw/DRAL16kHz/metadata/EN_transcripts.txt \
    data/raw/DRAL16kHz/metadata/ES_transcripts.txt
```

3. Synthesis experiment (long: roughly five hours on the test machine).

```
bvt run data/splits/dral_manifest_with_text.csv --split test \
    --directions en-es es-en \
    --output results/synthesis_test.csv --resume
python -m bilingual_voice.remeasure \
    results/synthesis_test.csv results/synthesis_final.csv --jobs 6
bvt summarize results/synthesis_final.csv \
    results/synthesis_summary.json
```

4. Corpus study and prosody prediction (feature extraction is long: about 25 minutes with six workers; the report runs in seconds and may be repeated freely).

```
python -m bilingual_voice.analysis features \
    data/splits/dral_manifest_with_text.csv \
    artifacts/features_wideband.csv --jobs 6
python -m bilingual_voice.analysis report \
    artifacts/features_wideband.csv artifacts/out_wideband \
    --features t1
```

5. Fine-tuning experiment (about 12 minutes per direction).

```
M=data/splits/dral_manifest_with_text.csv; O=artifacts/finetune
python -m bilingual_voice.finetune train $M $O --direction en-es
python -m bilingual_voice.finetune compare $M $O --direction en-es
```

6. Dissertation numbers and figures.

```
python thesis/scripts/thesis_numbers.py      # -> thesis/numbers.json
python thesis/scripts/thesis_figures.py      # -> thesis/figures/
./thesis/build.sh                             # -> thesis/main.pdf
```

7. Verification.

```
python -m pytest
python -m ruff check src tests webapp/backend
```

Every result row is checkpointed as it is produced, so an interrupted run resumes without repeating completed work; `--resume` skips completed keys but retries failures.

APPENDIX B

Full Prosody Prediction Results

Table 4.5 in Chapter 4 reports mean absolute error only. Table B.1 gives every statistic produced by the experiment, for all four systems. B1 predicts the training mean; B0 copies the source value; B2 adds a single global offset to B0; Ridge fits the residual from B0 with the regularisation strength selected on the development split.

Two entries look anomalous and are explained in Section 3.6.3. For the log duration ratio, B1 and B2 are mathematically identical because copy-source is zero, so no test between them is defined. And the errors of B0, B1 and B2 are symmetric between directions by construction; only the ridge rows genuinely differ.

Table B.1: Complete prosody prediction results on the held-out test split. MedAE is the median absolute error, reported as the robust statistic alongside the mean. p_{spk} is the Wilcoxon signed-rank test on per-speaker mean absolute errors ($n = 14$); “CI excl. 0” indicates whether the speaker-clustered bootstrap interval for the difference from B2 excludes zero.

Dir.	System	n	MAE	MedAE	RMSE	ΔB2	p_{spk}	CI excl. 0
<i>Target: F0 level (semitones)</i>								
EN $\rightarrow$ ES	B1	348	5.783	6.000	6.284	+327.4 %	0.0001	yes
EN $\rightarrow$ ES	B0	348	1.375	0.900	2.094	+1.6 %	0.0203	yes
EN $\rightarrow$ ES	B2	348	1.353	0.891	2.083	—	—	—
EN $\rightarrow$ ES	Ridge	348	1.353	0.892	2.083	−0.02 %	0.0040	yes
ES $\rightarrow$ EN	B1	347	5.947	6.275	6.445	+347.6 %	0.0001	yes
ES $\rightarrow$ EN	B0	347	1.350	0.896	2.028	+1.6 %	0.0238	yes
ES $\rightarrow$ EN	B2	347	1.329	0.889	2.017	—	—	—
ES $\rightarrow$ EN	Ridge	347	1.329	0.889	2.017	−0.00 %	0.9032	no
<i>Target: F0 range (semitones)</i>								
EN $\rightarrow$ ES	B1	348	1.034	0.848	1.417	+5.2 %	0.7148	no
EN $\rightarrow$ ES	B0	348	0.977	0.593	1.598	−0.6 %	0.1726	no
EN $\rightarrow$ ES	B2	348	0.983	0.581	1.606	—	—	—
EN $\rightarrow$ ES	Ridge	348	0.932	0.571	1.524	−5.14 %	0.0012	yes
ES $\rightarrow$ EN	B1	347	1.017	0.845	1.399	+5.0 %	0.5830	no
ES $\rightarrow$ EN	B0	347	0.963	0.588	1.569	−0.5 %	0.1726	no
ES $\rightarrow$ EN	B2	347	0.968	0.580	1.576	—	—	—
ES $\rightarrow$ EN	Ridge	347	0.931	0.558	1.502	−3.89 %	0.0001	yes
<i>Target: log duration ratio</i>								
EN $\rightarrow$ ES	B1	348	0.181	0.159	0.225	0.0 %	—	no
EN $\rightarrow$ ES	B0	348	0.186	0.161	0.229	+2.7 %	0.1189	yes
EN $\rightarrow$ ES	B2	348	0.181	0.159	0.225	—	—	—
EN $\rightarrow$ ES	Ridge	348	0.180	0.159	0.224	−0.73 %	0.1937	yes
ES $\rightarrow$ EN	B1	347	0.182	0.159	0.226	0.0 %	—	no
ES $\rightarrow$ EN	B0	347	0.187	0.161	0.229	+2.8 %	0.1040	yes
ES $\rightarrow$ EN	B2	347	0.182	0.159	0.226	—	—	—
ES $\rightarrow$ EN	Ridge	347	0.172	0.148	0.213	−5.34 %	0.1040	yes

Table B.2: Fitted ridge coefficients on standardised source features. The regularisation strength λ was selected on the development split by the one-standard-error rule. For F0 level the selected λ drives every weight to approximately zero, so the model reduces to the copy-plus-offset baseline.

Dir.	Target	λ	F0 level	F0 range	log dur.	voiced	words	rate
EN $\rightarrow$ ES	F0 level	10^6	-0.0006	-0.0007	-0.0001	-0.0001	0.0001	0.0002
EN $\rightarrow$ ES	F0 range	10^4	-0.0143	-0.1906	-0.0076	-0.0016	0.0006	0.0127
EN $\rightarrow$ ES	Duration ratio	10^4	-0.0006	-0.0023	-0.0056	0.0012	-0.0031	0.0055
ES $\rightarrow$ EN	F0 level	10^6	-0.0005	-0.0003	0.0001	-0.0001	0.0000	-0.0001
ES $\rightarrow$ EN	F0 range	10^4	0.0059	-0.1429	0.0003	-0.0115	0.0058	0.0073
ES $\rightarrow$ EN	Duration ratio	10^3	-0.0002	-0.0065	-0.0294	0.0127	0.0032	0.0394

APPENDIX C

Utterance-Level and Speaker-Level Statistics

Chapter 4 reports speaker means, because the test split ranges from 2 to 118 pairs per speaker and a mean over rows would be dominated by whoever was recorded most. Table C.1 gives both, so that the effect of the choice is visible.

Table C.1: Utterance-level and speaker-level means for the same measurements. The two agree closely except for English recognition error and Spanish-to-English intelligibility, where the most prolific speakers are not representative of the group.

Direction	Measure	Utterance mean	Speaker mean
EN → ES	Recognition WER	0.2087	0.1788
EN → ES	Synthesis intelligibility WER	0.1392	0.1352
EN → ES	Speaker similarity	0.4223	0.4206
EN → ES	Human ceiling	0.4365	0.4452
ES → EN	Recognition WER	0.2654	0.2634
ES → EN	Synthesis intelligibility WER	0.1348	0.1136
ES → EN	Speaker similarity	0.3888	0.3832
ES → EN	Human ceiling	0.4357	0.4450

Per speaker, identity as a percentage of the ceiling ranges from 74.7 % to 147.2 % in EN → ES (median 90.7 %) and from 55.5 % to 107.9 % in ES → EN (median 85.8 %). Values above 100 % arise where the synthesised output resembles the source recording more closely than the speaker’s own other-language recording does, which is possible because the synthesised audio inherits channel characteristics from the reference clip.

APPENDIX D

Supplementary Figures

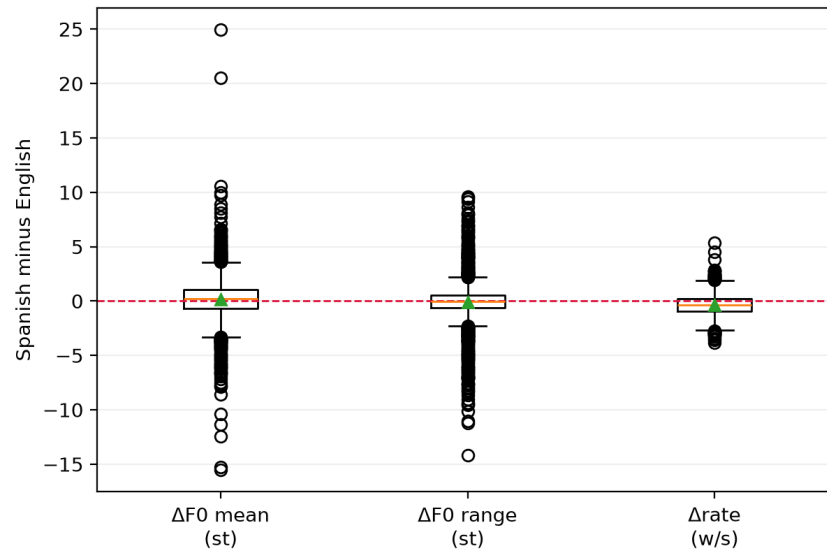


Figure D.1: Distribution of paired Spanish-minus-English differences for each prosodic measure over 2,304 same-speaker pairs. The systematic language-level shifts reported in Table 4.3 are small relative to the spread.

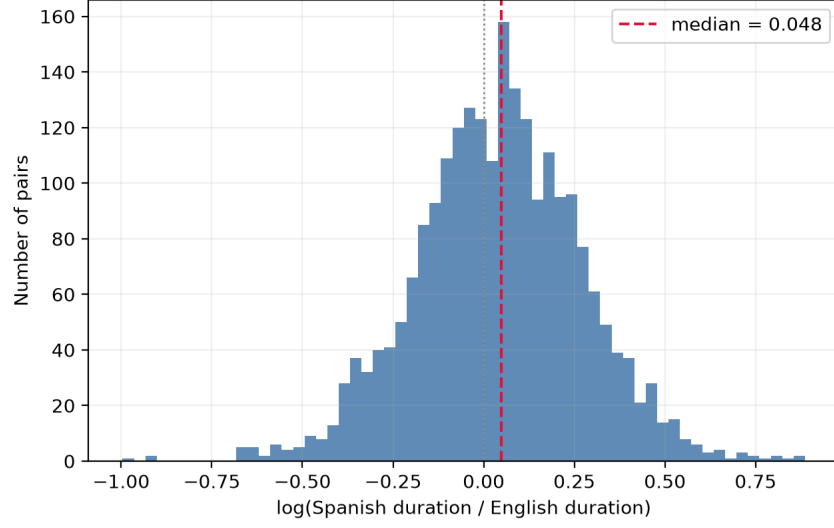


Figure D.2: Log ratio of Spanish to English duration for the same speaker and the same content. The median ratio of 1.049 is the human baseline against which the synthesiser’s $1.52\times$ inflation in Section 4.9 should be read.

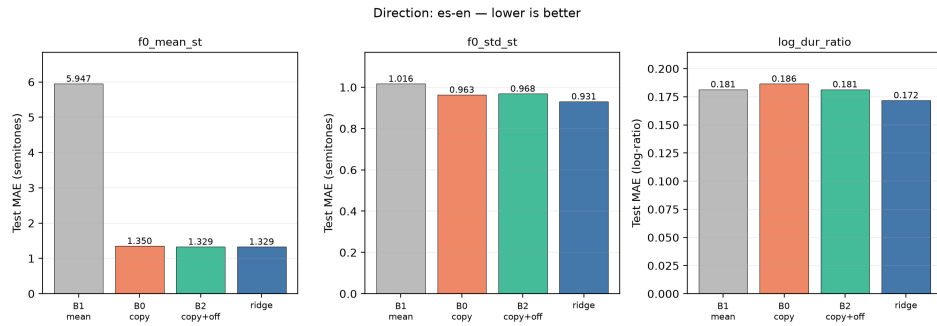


Figure D.3: Test mean absolute error for each prosody target in the ES $\rightarrow$ EN direction, model against the three baselines. The companion to Figure 4.6.

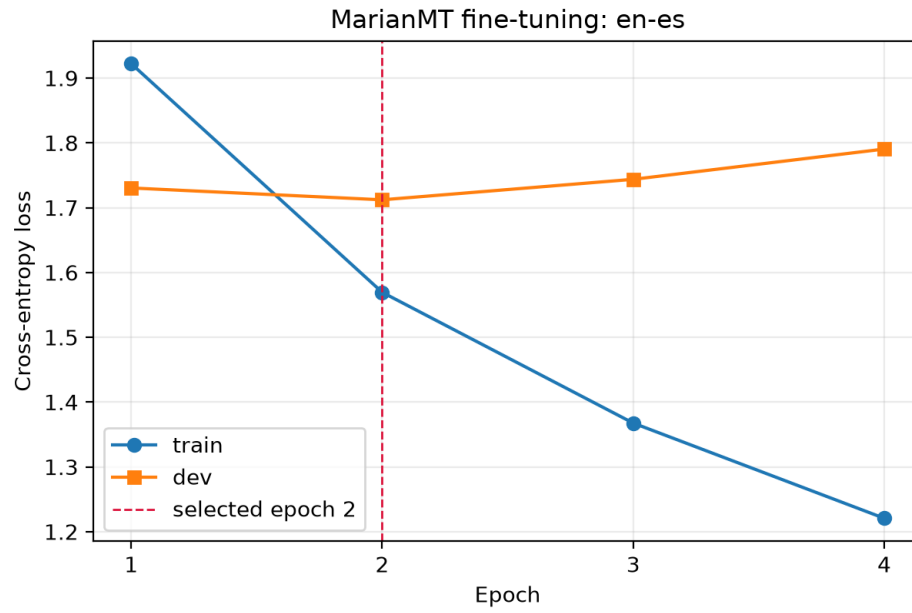


Figure D.4: Training and development loss per epoch for EN $\rightarrow$ ES fine-tuning. Development loss is minimised at epoch 2 and rises thereafter; the epoch-2 checkpoint was selected on that basis.

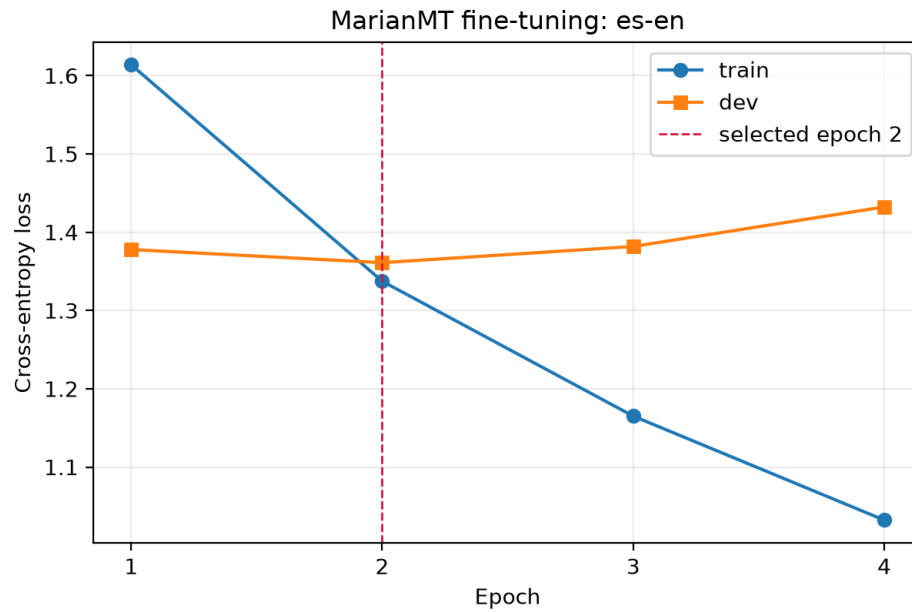


Figure D.5: Training and development loss per epoch for ES $\rightarrow$ EN fine-tuning. The same epoch was selected independently in this direction.

APPENDIX E

Computational Environment and Provenance

The record below is written automatically beside the analysis outputs each time the report is generated, so that the figures in this dissertation can be tied to an exact state of the code.

Item	Value
Git commit	ea7542c
Python	3.11.16
Platform	macOS 26.6, arm64 (Apple silicon)
NumPy / pandas / SciPy	2.4.6 / 2.3.3 / 1.17.1
librosa / matplotlib	0.11.0 / 3.11.1
Random seed	498 (splits, fine-tuning, bootstrap)
F0 search band	65–1000 Hz
F0 plausibility band	60–400 Hz
Quality gate	voiced ratio ≥ 0.30 ; duration 0.4–20 s
Feature set	t1: F0 level, F0 range, log duration, voiced ratio, word count, speaking rate
Regularisation grid	10^{-3} to 10^6 , ten values
Recognition model	Whisper small (CPU)
Translation models	Helsinki-NLP/opus-mt-en-es, opus-mt-es-en
Synthesis model	Coqui XTTS-v2 (CPU)
Speaker encoder	speechbrain/spkrec-ecapa-voxceleb (CPU)
Deep-learning framework	PyTorch (no TensorFlow or Keras is used anywhere)

Two components were adapted by the authors: the translation models, fine-tuned as described in Section 3.7, and the ridge prosody model, which is a closed-form statistical fit of six coefficients rather than a deep-learning model and is described as such throughout. The recognition, synthesis and speaker-embedding models are used exactly as released.